# UNIT - I

## Solution of Algebraic & Transcendental Equations

*(Bisection, Iteration, Regula-Falsi, Newton-Raphson, Jacobi & Gauss-Seidel)*

## 1. Introduction

*In engineering we often need the value of $x$ that satisfies $f(x)=0$. Such a value is called a ROOT (or ZERO) of the equation.*

- *Algebraic equation: a polynomial in $x$. e.g. $x^3-2x-5=0$, $x^2-3x+2=0$.*
- *Transcendental equation: contains trigonometric, exponential or logarithmic terms. e.g. $x\,e^x=2$, $\cos x=3x-1$, $x\log_{10}x=1.2$.*
- *Exact (analytical) formulas exist only for degree $\le 4$. For others we use NUMERICAL methods, which give an approximate root by successive approximation.*

### Basic facts to remember

- *Intermediate Value Theorem: if $f(x)$ is continuous on $[a,b]$ and $f(a)\cdot f(b)<0$, then at least one root lies in $(a,b)$.*
- *Initial approximation is found by TRIAL: tabulate $f(x)$ for $x=0,1,2,\ldots$ and look for a sign change.*
- *Descartes rule: number of +ve roots of a polynomial $\le$ number of sign changes in $f(x)$.*

### Order of convergence

*If $e_n=x_n-\alpha$ is the error at the $n$-th step and $|e_{n+1}|\approx k|e_n|^p$ then $p$ is the ORDER of convergence.*

| Method | Order p | Needs | Speed |
|---|---|---|---|
| Bisection | 1 (linear) | $f(a)f(b)<0$ | slow, sure |
| Iteration | 1 (linear) | $\lvert\varphi'(x)\rvert<1$ | slow |
| Regula-Falsi | 1 (about 1.0) | $f(a)f(b)<0$ | moderate |
| Newton-Raphson | 2 (quadratic) | $f'(x)\neq 0$ | very fast |

*Algorithm idea: start with a guess $\Rightarrow$ improve it by a formula $\Rightarrow$ repeat until two successive values agree to the required decimal places.*

## 2. Bisection Method (Bolzano / Interval-halving)

### Principle

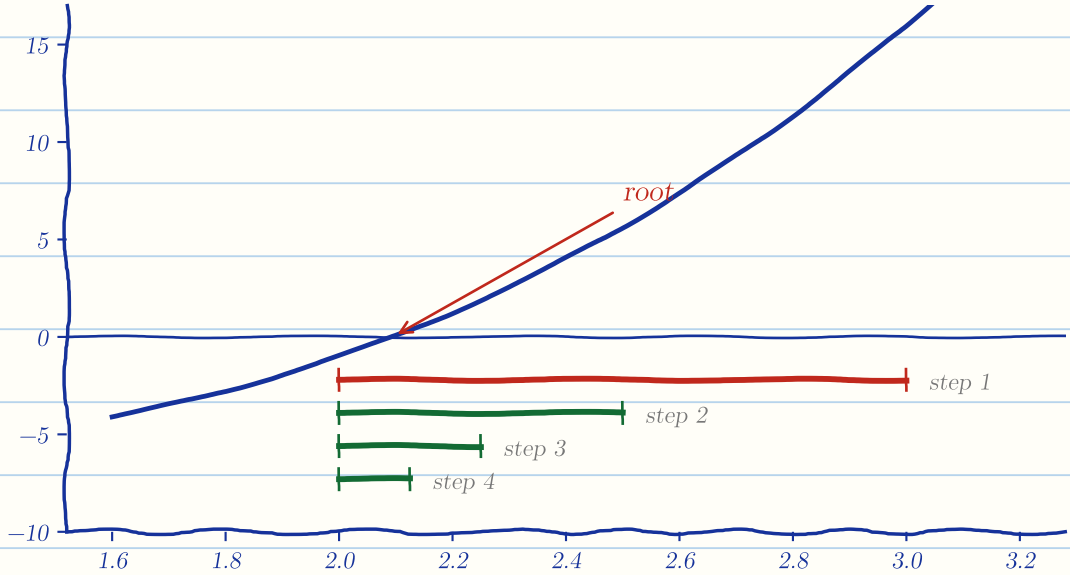
*Let $f$ be continuous on $[a,b]$ with $f(a)\,f(b)<0$. Then a root lies in $(a,b)$. Bisect the interval at the mid-point:*

$$x_0=\frac{a+b}{2}$$

- *If $f(x_0)=0$, then $x_0$ is the root.*
- *If $f(a)\,f(x_0)<0$, the root lies in $(a,x_0)$, so set $b=x_0$.*
- *Otherwise the root lies in $(x_0,b)$, so set $a=x_0$.*

*Repeat. After every step the interval length is HALVED.*

Bisection: interval halves each time




## Number of iterations needed

*After $n$ steps the length is $(b-a)/2^n$. For accuracy $\varepsilon$ :*

$$\frac{b-a}{2^n} < \varepsilon \;\Rightarrow\; n \geq \frac{\log(b-a) - \log\varepsilon}{\log 2}$$

*Merits: always converges, simple. Demerits: slow (linear), cannot find double roots / roots where $f$ does not change sign.*

**Example 1. Find a real root of $x^3 - 2x - 5 = 0$ by bisection (3 decimals)**

$f(2) = 8 - 4 - 5 = -1 < 0$ *and* $f(3) = 27 - 6 - 5 = 16 > 0$, *so a root lies in* $(2, 3)$.

| $n$ | $a$ (−ve) | $b$ (+ve) | $x_0 = (a+b)/2$ | $f(x_0)$ |
|---|---|---|---|---|
| 1 | 2.0000 | 3.0000 | 2.5000 | +5.6250 |
| 2 | 2.0000 | 2.5000 | 2.2500 | +1.8906 |
| 3 | 2.0000 | 2.2500 | 2.1250 | +0.3457 |
| 4 | 2.0000 | 2.1250 | 2.0625 | -0.3513 |
| 5 | 2.0625 | 2.1250 | 2.0938 | -0.0089 |
| 6 | 2.0938 | 2.1250 | 2.1094 | +0.1668 |

| $n$ | $a$ (−ve) | $b$ (+ve) | $x_0 = (a+b)/2$ | $f(x_0)$ |
|---|---|---|---|---|
| 7 | 2.0938 | 2.1094 | 2.1016 | +0.0786 |
| 8 | 2.0938 | 2.1016 | 2.0977 | +0.0347 |
| 9 | 2.0938 | 2.0977 | 2.0957 | +0.0129 |
| 10 | 2.0938 | 2.0957 | 2.0947 | +0.0020 |

*After 10 steps* $x \approx 2.0947$ *and* $f(x_0) \approx 0.002$, *so the root is* $x \approx 2.095$ *(true value 2.0946).*

**Example 2. Find the positive root of $x^3 - x - 1 = 0$ in (1,2)**

| $n$ | $a$ | $b$ | $x_0$ | $f(x_0)$ |
|---|---|---|---|---|
| 1 | 1.0000 | 2.0000 | 1.5000 | +0.8750 |
| 2 | 1.0000 | 1.5000 | 1.2500 | -0.2969 |
| 3 | 1.2500 | 1.5000 | 1.3750 | +0.2246 |
| 4 | 1.2500 | 1.3750 | 1.3125 | -0.0515 |
| 5 | 1.3125 | 1.3750 | 1.3438 | +0.0826 |
| 6 | 1.3125 | 1.3438 | 1.3281 | +0.0146 |
| 7 | 1.3125 | 1.3281 | 1.3203 | -0.0187 |

*Root* $\approx 1.32$ *(exact root 1.3247). Interval length after 7 steps = 1/128 = 0.0078.*

## 3. Iterative Method (Fixed-point)

*Rewrite* $f(x) = 0$ *in the form* $x = \varphi(x)$. *Start with* $x_0$ *and compute*

$$x_{n+1} = \varphi(x_n), \quad n = 0, 1, 2, \ldots$$

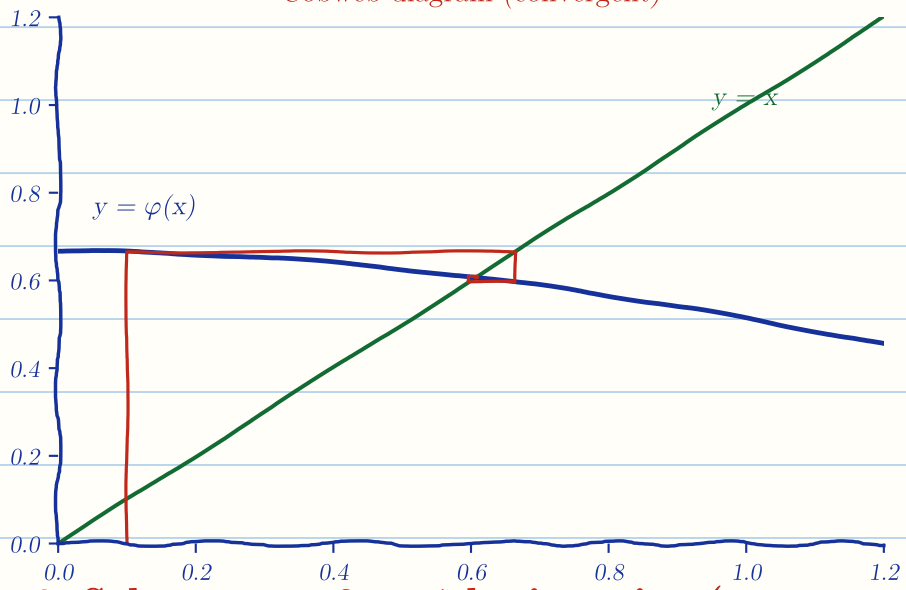
**Key condition**

*Condition for convergence:* $|\varphi'(x)| < 1$ *for all* $x$ *in the interval containing the root.*

*Smaller* $|\varphi'| \Rightarrow$ *faster convergence. Error:* $e_{n+1} \approx \varphi'(\alpha)\, e_n$ *(linear).*

*Choosing* $\varphi$*: write the equation in several ways, pick the one with* $|\varphi'(x)| < 1$ *near the root. e.g. for* $x^3 + x - 1 = 0$ *take* $x = (1+x)^{-1/2}$ *or* $x = 1 - x^3$ *(check derivative!).*

*Cobweb diagram (convergent)*




**Example 3. Solve** $\cos x = 3x - 1$ **by iteration (start** $x_0 = 0.5$**)**

*Write* $x = \varphi(x) = \dfrac{1+\cos x}{3}$ , $\varphi'(x) = -\dfrac{\sin x}{3}$ , $|\varphi'| \le \frac{1}{3} < 1 \Rightarrow$ *convergent.*

| $n$ | $x_n$ | $x_{n+1} = (1+\cos x_n)/3$ |
|---|---|---|
| 0 | 0.50000 | 0.62586 |
| 1 | 0.62586 | 0.60349 |
| 2 | 0.60349 | 0.60779 |
| 3 | 0.60779 | 0.60697 |
| 4 | 0.60697 | 0.60713 |

| $n$ | $x_n$ | $x_{n+1} = (1+\cos x_n)/3$ |
|---|---|---|
| 5 | 0.60713 | 0.60710 |
| 6 | 0.60710 | 0.60710 |

*Root* $\approx 0.60710$ *(correct to 4 decimals; cos taken in RADIANS).*

**Example 4. Find a root of $2x - \log_{10} x = 7$ (near 3.8)**

$x = \varphi(x) = \frac{1}{2}(\log_{10} x + 7)$, $\varphi'(x) = \frac{0.4343}{2x}$ *which is* $< 1$ *for* $x \approx 3.8$.

| $n$ | $x_n$ | $x_{n+1}$ |
|---|---|---|
| 0 | 3.80000 | 3.78989 |
| 1 | 3.78989 | 3.78931 |
| 2 | 3.78931 | 3.78928 |
| 3 | 3.78928 | 3.78928 |
| 4 | 3.78928 | 3.78928 |

*Hence the root is* $x \approx 3.78893$.

## 4. Regula-Falsi Method (Method of False Position)

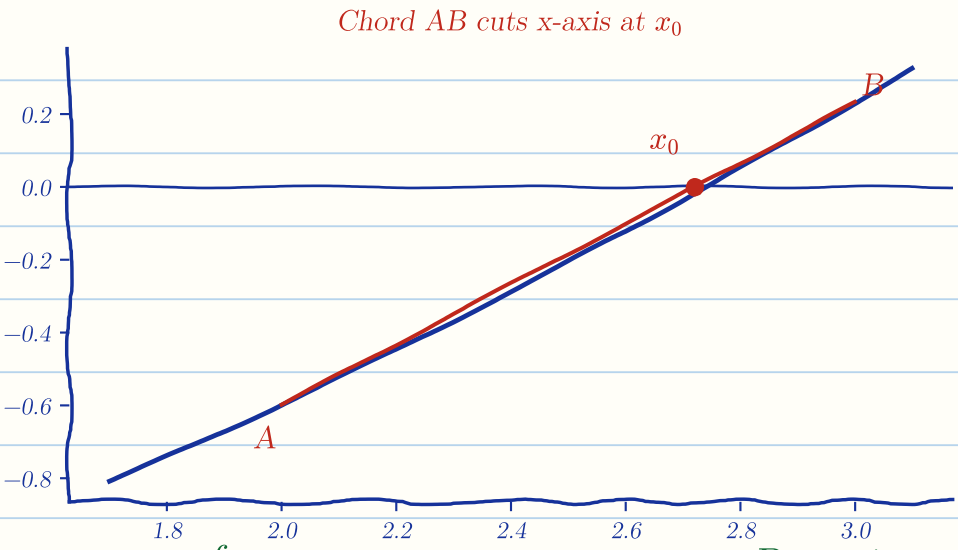
*Replace the curve* $y = f(x)$ *on* $[a, b]$ *by the CHORD joining* $A(a, f(a))$ *and* $B(b, f(b))$ *. The point where the chord cuts the* $x$*-axis is the next approximation.*

*Equation of chord:* $y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a)$*. Put* $y = 0$*:*

**Regula-Falsi formula**

$$x_0 = \frac{a\,f(b) - b\,f(a)}{f(b) - f(a)} = a - \frac{f(a)}{f(b) - f(a)}\,(b - a)$$

- *If* $f(a)f(x_0) < 0$ *the root is in* $(a, x_0)$*: replace* $b$ *by* $x_0$*. Otherwise replace* $a$ *by* $x_0$*.*
- *Repeat, always using the NEW end-points in the formula, until successive values agree.*




*Merits: converges for sure, no derivative needed. Demerit: one end-point may stay fixed, so convergence is slow (order $\approx 1$).*

**Example 5. Find a root of $x\log_{10} x = 1.2$ by Regula-Falsi**

$f(2) = -0.5979 < 0,\ f(3) = +0.2314 > 0$ , *root in* $(2,3)$. *With* $a = 2,\ b = 3$:

| $n$ | $a$ | $b$ | $x_0$ | $f(x_0)$ |
|---|---|---|---|---|
| 1 | 2.0000 | 3.0000 | 2.72101 | -0.01709 |
| 2 | 2.7210 | 3.0000 | 2.74021 | -0.00038 |
| 3 | 2.7402 | 3.0000 | 2.74064 | -0.00001 |
| 4 | 2.7406 | 3.0000 | 2.74065 | -0.00000 |
| 5 | 2.7406 | 3.0000 | 2.74065 | -0.00000 |

*Root* $\approx 2.7406$ *(correct to 4 decimals).*

**Example 6. Find the root of $x^3 - 2x - 5 = 0$ by Regula-Falsi (in (2,3))**

| $n$ | $a$ | $b$ | $x_0$ | $f(x_0)$ |
|---|---|---|---|---|
| 1 | 2.0000 | 3.0000 | 2.05882 | -0.39080 |
| 2 | 2.0588 | 3.0000 | 2.08126 | -0.14720 |

| $n$ | $a$ | $b$ | $x_0$ | $f(x_0)$ |
|---|---|---|---|---|
| 3 | 2.0813 | 3.0000 | 2.08964 | -0.05468 |
| 4 | 2.0896 | 3.0000 | 2.09274 | -0.02020 |
| 5 | 2.0927 | 3.0000 | 2.09388 | -0.00745 |

*Compare: bisection took 10 steps for 3 decimals; here the root 2.0946 is reached much sooner.*

## 5. Newton-Raphson Method

*Let $x_0$ be an approximate root and $x_1 = x_0 + h$ the better value, so $f(x_0 + h) \approx 0$. Taylor series (first two terms):*

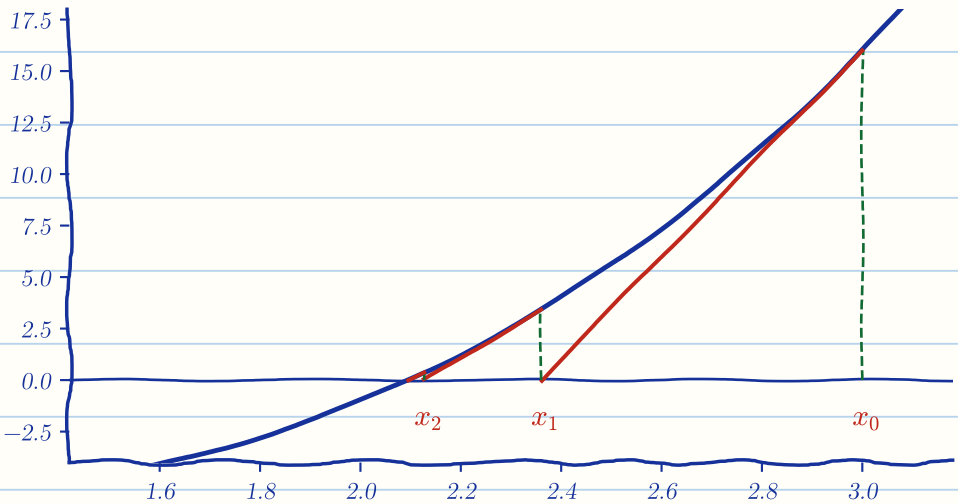
$$f(x_0) + h\, f'(x_0) = 0 \;\Rightarrow\; h = -\frac{f(x_0)}{f'(x_0)}$$

**Newton-Raphson formula**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \qquad n = 0, 1, 2, \ldots$$

*Geometrically: draw the TANGENT at $(x_n, f(x_n))$; the point where it meets the $x$-axis is $x_{n+1}$.*

Tangents converge quickly to the root




### Convergence & conditions

- *Order of convergence = 2 (quadratic): the number of correct decimals roughly DOUBLES every step.*

- *Convergence is guaranteed if* $|f(x)\,f''(x)| < |f'(x)|^2$ *in the interval around the root.*
- *Fails (or is slow) when* $f'(x_n) \approx 0$*; choose* $x_0$ *close to the root. Equivalent to iteration with* $\varphi(x) = x - f/f'$.

**Example 7. Find a root of $x^3 - 2x - 5 = 0$ (take $x_0 = 2$)**

$f(x) = x^3 - 2x - 5, f'(x) = 3x^2 - 2.$

| $n$ | $x_n$ | $f(x_n)$ | $f'(x_n)$ | $x_{n+1}$ |
|---|---|---|---|---|
| 0 | 2.00000 | -1.00000 | 10.00000 | 2.10000 |
| 1 | 2.10000 | 0.06100 | 11.23000 | 2.09457 |
| 2 | 2.09457 | 0.00019 | 11.16165 | 2.09455 |
| 3 | 2.09455 | 0.00000 | 11.16144 | 2.09455 |

*Root* $\approx 2.09455$. *Only 3 iterations needed!*

**Example 8. Find a real root of $x\,e^x = 2$ (take $x_0 = 1$)**

$f(x) = xe^x - 2,\ \ f'(x) = e^x(1 + x).$

| $n$ | $x_n$ | $f(x_n)$ | $f'(x_n)$ | $x_{n+1}$ |
|---|---|---|---|---|
| 0 | 1.00000 | 0.71828 | 5.43656 | 0.86788 |
| 1 | 0.86788 | 0.06716 | 4.44902 | 0.85278 |
| 2 | 0.85278 | 0.00077 | 4.34694 | 0.85261 |
| 3 | 0.85261 | 0.00000 | 4.34575 | 0.85261 |

*Root* $\approx 0.85261$.

**Important deductions from Newton-Raphson formula**

*(i) Square root of* $N$ $(f = x^2 - N)$*:*

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)$$

*(ii) Reciprocal of* $N$ $(f = \frac{1}{x} - N)$*:* $x_{n+1} = x_n\,(2 - N x_n)$

*(iii) k-th root of N* $(f = x^k - N)$: $x_{n+1} = \frac{1}{k}\left[(k-1)x_n + \frac{N}{x_n^{k-1}}\right]$

**Example 9. Find $\sqrt{12}$ using Newton-Raphson ($x_0 = 3.5$)**

| $n$ | $x_n$ | $x_{n+1} = \frac{1}{2}(x_n + 12/x_n)$ |
|---|---|---|
| 0 | 3.50000 | 3.46429 |
| 1 | 3.46429 | 3.46410 |
| 2 | 3.46410 | 3.46410 |

$\sqrt{12} = 3.46410$. *Check:* $3.4641^2 = 11.99999$.

**Example 10. Find the reciprocal of 18 ($x_0 = 0.05$)**

| $n$ | $x_n$ | $x_{n+1} = x_n(2 - 18x_n)$ |
|---|---|---|
| 0 | 0.050000 | 0.055000 |
| 1 | 0.055000 | 0.055550 |
| 2 | 0.055550 | 0.055556 |

$1/18 = 0.055555..$ *correct.*

## 6. System of Linear Algebraic Equations: Jacobi & Gauss-Seidel

*For large systems direct methods (Cramer, Gauss elimination) are lengthy; ITERATIVE methods are used. Consider*

$$a_1x + b_1y + c_1z = d_1, \quad a_2x + b_2y + c_2z = d_2, \quad a_3x + b_3y + c_3z = d_3$$

**Convergence condition**

*Condition: the system must be DIAGONALLY DOMINANT, i.e.*

$$|a_1| > |b_1| + |c_1|, \quad |b_2| > |a_2| + |c_2|, \quad |c_3| > |a_3| + |b_3|$$

*If not, interchange the equations first to make it so.*

*Solve for the diagonal unknowns:*

$$x=\frac{1}{a_1}(d_1-b_1y-c_1z),\quad y=\frac{1}{b_2}(d_2-a_2x-c_2z),\quad z=\frac{1}{c_3}(d_3-a_3x-b_3y)$$

## (a) Jacobi iteration (method of simultaneous displacement)

*Start with* $x^{(0)}=y^{(0)}=z^{(0)}=0$ *(or any guess). Compute ALL new values using only the OLD values:*

$$x^{(k+1)}=\frac{1}{a_1}(d_1-b_1y^{(k)}-c_1z^{(k)}),\quad y^{(k+1)}=\frac{1}{b_2}(d_2-a_2x^{(k)}-c_2z^{(k)})$$

$$z^{(k+1)}=\frac{1}{c_3}(d_3-a_3x^{(k)}-b_3y^{(k)})$$

## (b) Gauss-Seidel iteration (method of successive displacement)

*Use the LATEST available value immediately in the next equation:*

$$x^{(k+1)}=\frac{1}{a_1}(d_1-b_1y^{(k)}-c_1z^{(k)})$$

$$y^{(k+1)}=\frac{1}{b_2}(d_2-a_2x^{(k+1)}-c_2z^{(k)}),\quad z^{(k+1)}=\frac{1}{c_3}(d_3-a_3x^{(k+1)}-b_3y^{(k+1)})$$

*Gauss-Seidel converges about TWICE as fast as Jacobi and needs less storage.*

**Example 11. Solve:**

$10x+y+z=12,\quad 2x+10y+z=13,\quad 2x+2y+10z=14$

*Diagonally dominant (\$10>1+1,\ 10>2+1,\ 10>2+2\$). Rewrite:*

$x=\frac{1}{10}(12-y-z),\ y=\frac{1}{10}(13-2x-z),\ z=\frac{1}{10}(14-2x-2y).$

**Jacobi (start 0,0,0):**

| Iteration | $x$ | $y$ | $z$ |
|---|---|---|---|
| 1 | 1.2000 | 1.3000 | 1.4000 |
| 2 | 0.9300 | 0.9200 | 0.9000 |

| Iteration | $x$ | $y$ | $z$ |
|---|---|---|---|
| 3 | 1.0180 | 1.0240 | 1.0300 |
| 4 | 0.9946 | 0.9934 | 0.9916 |
| 5 | 1.0015 | 1.0019 | 1.0024 |
| 6 | 0.9996 | 0.9995 | 0.9993 |

**Gauss-Seidel (start 0,0,0):**

| Iteration | $x$ | $y$ | $z$ |
|---|---|---|---|
| 1 | 1.2000 | 1.0600 | 0.9480 |
| 2 | 0.9992 | 1.0054 | 0.9991 |
| 3 | 0.9996 | 1.0002 | 1.0001 |
| 4 | 1.0000 | 1.0000 | 1.0000 |
| 5 | 1.0000 | 1.0000 | 1.0000 |

*Solution:* $x=1,\ y=1,\ z=1$. *Gauss-Seidel is correct to 4 decimals by the 5th step; Jacobi needs a few more.*

**Example 12. Solve by Gauss-Seidel:**

$20x+y-2z=17,\ 3x+20y-z=-18,\ 2x-3y+20z=25$

$x=\frac{1}{20}(17-y+2z),\ y=\frac{1}{20}(-18-3x+z),\ z=\frac{1}{20}(25-2x+3y).$

| Iteration | $x$ | $y$ | $z$ |
|---|---|---|---|
| 1 | 0.8500 | -1.0275 | 1.0109 |
| 2 | 1.0025 | -0.9998 | 0.9998 |
| 3 | 1.0000 | -1.0000 | 1.0000 |
| 4 | 1.0000 | -1.0000 | 1.0000 |

*Solution* $x=1,\ y=-1,\ z=1.$

## 7. Quick Revision Sheet

**Formulas at a glance**

*Bisection :* $x_0 = \frac{a+b}{2}$ ; $n \geq \frac{\log(b-a) - \log \varepsilon}{\log 2}$

Regula-Falsi: $x_0 = \dfrac{af(b) - bf(a)}{f(b) - f(a)}$ Newton: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

*Iteration :* $x_{n+1} = \varphi(x_n)$, *needs* $|\varphi'(x)| < 1$

*Jacobi: use old values ; Gauss-Seidel: use newest values (faster).*

*Both need diagonal dominance.*

- *Always check the sign change* $f(a)f(b) < 0$ *first and write* $f(a), f(b)$ *explicitly in the exam.*
- *Keep one extra decimal in intermediate steps; stop when two consecutive iterates agree to the asked decimals.*
- *Calculator in RADIAN mode for trigonometric equations.*
- *Expected exam questions: bisection/regula-falsi/N-R problem (5 marks each), Jacobi or Gauss-Seidel problem (5 marks), derivation of N-R formula (short answer).*

# UNIT - II

## Interpolation & Curve Fitting

*(Finite differences, Newton forward / backward, Lagrange, Least squares)*

## 1. Introduction to Interpolation

*Suppose values of a function $y = f(x)$ are known only at a few points $x_0, x_1, \ldots, x_n$ (a table of data). The process of finding the value of $y$ at a point INSIDE the range of the table is INTERPOLATION; outside the range it is EXTRAPOLATION.*

- *Equal spacing ($x_{i+1} - x_i = h$): use Newton forward / backward formulae.*
- *Unequal spacing: use Lagrange's formula (or divided differences).*
- *The interpolating polynomial $P_n(x)$ of degree $\le n$ passes through all $n+1$ given points and is UNIQUE.*

## 2. Finite Differences

### Forward differences ($\Delta$)

*Let $y_0, y_1, \ldots$ be the values at $x_0,\ x_0 + h,\ x_0 + 2h, \ldots$ Then*

$$\Delta y_0 = y_1 - y_0, \quad \Delta y_1 = y_2 - y_1, \quad \ldots \quad \Delta y_r = y_{r+1} - y_r$$

$$\Delta^2 y_r = \Delta y_{r+1} - \Delta y_r, \qquad \Delta^n y_r = \Delta^{n-1} y_{r+1} - \Delta^{n-1} y_r$$

### Backward differences ($\nabla$)

$$\nabla y_r = y_r - y_{r-1}, \qquad \nabla^2 y_r = \nabla y_r - \nabla y_{r-1}$$

*Note: $\nabla y_1 = \Delta y_0$, $\nabla^2 y_2 = \Delta^2 y_0$ - the same numbers appear in a diagonal-vs-horizontal position of the table.*

## Central difference ($\delta$) and shift operator ($E$), averaging ($\mu$)

$$\delta y_r = y_{r+\frac{1}{2}} - y_{r-\frac{1}{2}}, \qquad E y_r = y_{r+1}, \qquad E^n y_r = y_{r+n}$$

**Operator relations**

$$\Delta = E - 1, \qquad \nabla = 1 - E^{-1}, \qquad \delta = E^{1/2} - E^{-1/2}$$

$$E = 1 + \Delta, \qquad \nabla = \Delta E^{-1}, \qquad \mu = \frac{1}{2}(E^{1/2} + E^{-1/2})$$

*Also:* $\Delta\nabla = \nabla\Delta = \delta^2$ , $E\nabla = \Delta$ , $(1+\Delta)(1-\nabla) = 1$.

### Important results

- $\Delta^n$ *of a polynomial of degree* $n$ *is a CONSTANT equal to* $n!\, a_0\, h^n$ ; $\Delta^{n+1} = 0$.
- $\Delta(c) = 0$, $\Delta(af + bg) = a\Delta f + b\Delta g$ *(linear operator).*
- $\Delta^n y_0 = \sum_{r=0}^{n} (-1)^r \binom{n}{r} y_{n-r}$ , *e.g.* $\Delta^2 y_0 = y_2 - 2y_1 + y_0$, $\Delta^3 y_0 = y_3 - 3y_2 + 3y_1 - y_0$.
- $y_n = E^n y_0 = (1+\Delta)^n y_0 = y_0 + n\Delta y_0 + \binom{n}{2}\Delta^2 y_0 + \ldots$

### Example 1. Forward difference table for $y = x^3 + x$ (x = 0,1,2,3,4,5)

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|---|---|---|---|---|---|
| 0 | 0 | 2 | 6 | 6 | 0 |
| 1 | 2 | 8 | 12 | 6 | 0 |
| 2 | 10 | 20 | 18 | 6 | |
| 3 | 30 | 38 | 24 | | |
| 4 | 68 | 62 | | | |

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|---|---|---|---|---|---|
| 5 | 130 | | | | |

$\Delta^3 y = 6 = constant = 3! \times 1 \times 1^3$ (degree 3, $a_0 = 1$), and $\Delta^4 y = 0$.

## 3. Newton's Forward Interpolation Formula

*Used when the point lies NEAR THE BEGINNING of the table.*

*Take* $x = x_0 + uh$, *i.e.* $u = \dfrac{x - x_0}{h}$.

**Newton-Gregory forward formula**

$$y(x) = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots$$

*Terms continue until the differences vanish / table ends.*

*Derivation (sketch):* since $y_u = E^u y_0 = (1+\Delta)^u y_0$, expand by the binomial theorem: $(1+\Delta)^u = 1 + u\Delta + \frac{u(u-1)}{2!}\Delta^2 + \cdots$

**Example 2. Population data: find the population in 1895**

| **Year** $x$ | **Pop.** $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|---|---|---|---|---|---|
| 1891 | 46 | 20 | -5 | 2 | -3 |
| 1901 | 66 | 15 | -3 | -1 | |
| 1911 | 81 | 12 | -4 | | |
| 1921 | 93 | 8 | | | |
| 1931 | 101 | | | | |

$$x = 1895,\ x_0 = 1891,\ h = 10 \ \Rightarrow\ u = \frac{1895 - 1891}{10} = 0.4$$

$$y(1895) = 46 + (0.4)(20) + \frac{(0.4)(-0.6)}{2}(-5) + \frac{(0.4)(-0.6)(-1.6)}{6}(2)$$

$$+ \frac{(0.4)(-0.6)(-1.6)(-2.6)}{24}(-3)$$

$$= 46 + 8 + 0.60 + 0.128 + 0.1248 \Rightarrow y(1895) \approx 54.8528 \text{ (about } 54.85$$

*thousands).*

**Example 3. Given sin 45 = 0.7071, sin 50 = 0.7660, sin 55 = 0.8192, sin 60 = 0.8660; find sin 52**

| $x^\circ$ | $y = \sin x$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ |
|---|---|---|---|---|
| 45 | 0.7071 | 0.0589 | -0.0057 | -0.0007 |
| 50 | 0.7660 | 0.0532 | -0.0064 | |
| 55 | 0.8192 | 0.0468 | | |
| 60 | 0.8660 | | | |

$u = \dfrac{52-45}{5} = 1.4.$

$\sin 52^\circ = 0.7071 + 1.4(0.0589) + \dfrac{1.4(0.4)}{2}(-0.0057) + \dfrac{1.4(0.4)(-0.6)}{6}(-0.000$

$\Rightarrow \sin 52^\circ \approx 0.7880$ *(table value 0.7880).*

## 4. Newton's Backward Interpolation Formula

*Used when the point lies NEAR THE END of the table. Take the last point* $x_n$ *and* $v = \dfrac{x - x_n}{h}$ *(negative).*

**Newton-Gregory backward formula**

$$y(x) = y_n + v\,\nabla y_n + \frac{v(v+1)}{2!}\nabla^2 y_n + \frac{v(v+1)(v+2)}{3!}\nabla^3 y_n + \cdots$$

*(Derived from* $y_v = E^v y_n = (1-\nabla)^{-v} y_n$*.) Differences used are those along the LAST ROW (diagonal) of the table.*

**Example 4. Using the data of Example 3 find sin 57**

$v = \dfrac{57-60}{5} = -0.6$ ; $\nabla y_n = 0.0468$, $\nabla^2 y_n = -0.0064$, $\nabla^3 y_n = -0.0007$

*(last entries of each column).*

$\sin 57^\circ = 0.8660 + (-0.6)(0.0468) + \dfrac{(-0.6)(0.4)}{2}(-0.0064) + \dfrac{(-0.6)(0.4)(1.4}{6}$

*(table 0.8387).*

## Example 5. Same population table: find population in 1925

$v = \dfrac{1925 - 1931}{10} = -0.6;$

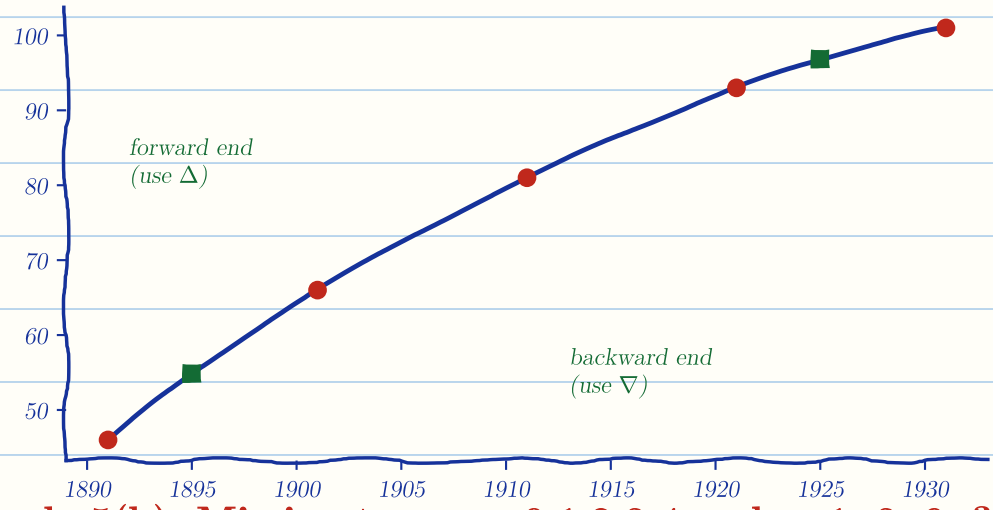
$\nabla y_n = 8,\ \nabla^2 y_n = -4,\ \nabla^3 y_n = -1,\ \nabla^4 y_n = -3.$ *Then*

$$y(1925) = 101 + (-0.6)(8) + \frac{(-0.6)(0.4)}{2}(-4) + \frac{(-0.6)(0.4)(1.4)}{6}(-1)$$

$$+ \frac{(-0.6)(0.4)(1.4)(2.4)}{24}(-3)$$

$y(1925) \approx 96.8368$ *(about 96.84 thousands).*

*Forward formula near start, backward near end*




## Example 5(b). Missing term: x: 0,1,2,3,4 and y: 1, 3, 9, ?, 81 - find the missing value

*Four values are known, so the polynomial has degree* $\leq 3$ *and hence* $\Delta^4 y_0 = 0$*:*

$$y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0 \Rightarrow 81 - 4y_3 + 54 - 12 + 1 = 0 \Rightarrow y_3 = 31$$

*Rule: with* $k$ *known values,* $\Delta^k y = 0$ *(assume polynomial of degree* $k-1$*).*

## Error in interpolation

*If* $f$ *is* $(n+1)$ *times differentiable, the error of the* $n$*-th degree interpolating polynomial is*

$$E(x) = f(x) - P_n(x) = \frac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!} f^{(n+1)}(\xi), \quad \xi \in (x_0, x_n)$$

*The error is smallest near the middle of the table and largest near the ends; extrapolation is therefore unreliable.*

## 5. Lagrange's Interpolation Formula (unequal intervals)

*Given $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$ with ARBITRARY spacing, the polynomial of degree $n$ through them is*

**Lagrange's formula**

$$y = f(x) = \sum_{i=0}^{n} y_i \prod \frac{x - x_j}{x_i - x_j}$$

$$= \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)} y_1 + \cdots$$

- *For 3 points $(x_0, x_1, x_2)$ there are three terms; each term omits its own factor in numerator & denominator.*
- *INVERSE interpolation: interchange the roles of $x$ and $y$ in the formula to find $x$ for a given $y$.*
- *No need for equal spacing and no difference table; but adding a new point means recomputing everything.*

**Example 6. Given (5,12), (6,13), (9,14), (11,16) find $y(10)$**

*At $x = 10$: numerator $\prod(x - x_j)$ and denominator $\prod(x_i - x_j)$ (over $j \neq i$):*

| Point | Numerator | Denominator | $y_i$ | Term |
|---|---|---|---|---|
| $x_0$=5 | -4 | -24 | 12 | 2.0000 |
| $x_1$=6 | -5 | 15 | 13 | -4.3333 |
| $x_2$=9 | -20 | -24 | 14 | 11.6667 |
| $x_3$=11 | 20 | 60 | 16 | 5.3333 |

$y(10) = 14.6667 \approx 14.6667$ .

**Example 6(b). Inverse interpolation: for the same data find $x$ when $y = 15$**

*Interchange* $x \leftrightarrow y$: $x = \sum x_i \prod_{j \neq i} \frac{y - y_j}{y_i - y_j}$ , $y = 15$ .

*Weights* $L_0 = 0.2500,\ L_1 = -1.0000,\ L_2 = 1.5000,\ L_3 = 0.2500 \Rightarrow$ $x \approx 11.5000$.

**Example 7. Find the polynomial through (0,2), (1,3), (2,12), (5,147)**

*Using Lagrange's formula and simplifying term by term (or by differences), we get*

$$f(x) = x^3 + x^2 - x + 2$$

*Check:* $f(1) = 1 + 1 - 1 + 2 = 3$ ; $f(2) = 8 + 4 - 2 + 2 = 12$ ; $f(5) = 125 + 25 - 5 + 2 = 147$ . *Also* $f(3) = 27 + 9 - 3 + 2 = 35$.

*Verified numerically: coefficients = 1.0, 1.0, -1.0, 2.0*

## 6. Curve Fitting - Method of Least Squares

*Given $n$ data points $(x_i, y_i)$ we want the curve $y = f(x)$ that fits the trend. Let $e_i = y_i - f(x_i)$ be the residuals. The LEAST SQUARES method chooses the constants to minimise*

$$E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - f(x_i)]^2$$

*Setting* $\frac{\partial E}{\partial a} = 0$, $\frac{\partial E}{\partial b} = 0$, … *gives the NORMAL EQUATIONS.*

### (a) Straight line $y = a + bx$

**Normal equations (2 unknowns a, b)**

$$\sum y = na + b\sum x$$
$$\sum xy = a\sum x + b\sum x^2$$

*Solving:* $b = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$ , $a = \bar{y} - b\bar{x}$.

**Example 8. Fit a straight line to x: 1,2,3,4,5 and y: 14,27,40,55,68**

| $x$ | $y$ | $xy$ | $x^2$ |
|---|---|---|---|
| 1 | 14 | 14 | 1 |
| 2 | 27 | 54 | 4 |
| 3 | 40 | 120 | 9 |
| 4 | 55 | 220 | 16 |
| 5 | 68 | 340 | 25 |
| 15 | 204 | 748 | 55 |

*Last row = sums:* $\sum x = 15$, $\sum y = 204$, $\sum xy = 748$, $\sum x^2 = 55$,

$n = 5$.

$204 = 5a + 15b$ , $748 = 15a + 55b \Rightarrow b = 13.6$, $a = -0.0$.

$$y = -0.0 + 13.6\,x$$

### (b) Second-degree parabola $y = a + bx + cx^2$

**Normal equations (3 unknowns a, b, c)**

$$\sum y = na + b\sum x + c\sum x^2$$
$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$
$$\sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

*SHORT-CUT: if $x$ values are equally spaced, shift the origin ($u = (x - \text{mid})/h$) so that $\sum u = \sum u^3 = 0$; the equations simplify greatly.*

**Example 9. Fit $y = a + bx + cx^2$ to x: 0,1,2,3,4 and y: 1, 1.8, 1.3, 2.5, 6.3**

| $x$ | $y$ | $x^2$ | $x^3$ | $x^4$ | $xy$ | $x^2y$ |
|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1.8 | 1 | 1 | 1 | 1.8 | 1.8 |
| 2 | 1.3 | 4 | 8 | 16 | 2.6 | 5.2 |
| 3 | 2.5 | 9 | 27 | 81 | 7.5 | 22.5 |
| 4 | 6.3 | 16 | 64 | 256 | 25.2 | 100.8 |
| 10 | 12.9 | 30 | 100 | 354 | 37.1 | 130.3 |

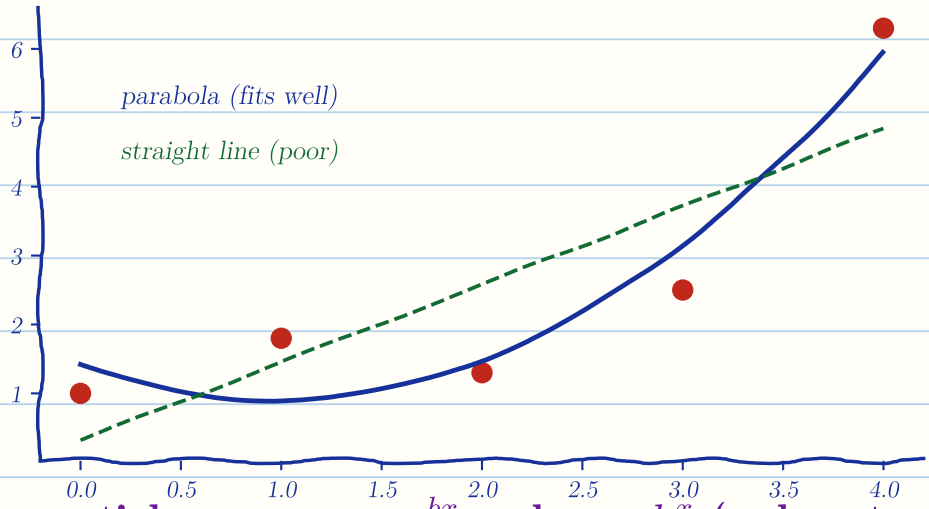
$12.9 = 5a + 10b + 30c$ ; $37.1 = 10a + 30b + 100c$ ;

$130.3 = 30a + 100b + 354c$ .

*Solving:* $a = 1.420,\ b = -1.070,\ c = 0.550.$

$$y = 1.42 - 1.07\,x + 0.55\,x^2$$

*Least-squares fits to Example 9*




## (c) Exponential curves $y = ae^{bx}$ and $y = ab^x$ (reduce to a line)

*Take natural log:* $y = ae^{bx} \Rightarrow \ln y = \ln a + bx$. *Put* $Y = \ln y$, $A = \ln a$: *line* $Y = A + bx$. *Normal equations:*

$$\sum Y = nA + b\sum x, \qquad \sum xY = A\sum x + b\sum x^2$$

*Then* $a = e^A$. *For* $y = ab^x$: $\log y = \log a + x\log b$ (*use* $\log_{10}$, *then* $a = 10^A$, $b = 10^B$). *For power curve* $y = ax^b$ : $\log y = \log a + b\log x$.

**Example 10. Fit $y = ae^{bx}$ to x: 1,2,3,4,5,6 ; y: 7.209, 5.265, 3.846, 2.809, 2.052, 1.499**

| $x$ | $y$ | $Y = \ln y$ | $xY$ | $x^2$ |
|---|---|---|---|---|
| 1 | 7.209 | 1.9753 | 1.9753 | 1 |
| 2 | 5.265 | 1.6611 | 3.3222 | 4 |
| 3 | 3.846 | 1.3470 | 4.0411 | 9 |
| 4 | 2.809 | 1.0328 | 4.1313 | 16 |
| 5 | 2.052 | 0.7188 | 3.5941 | 25 |
| 6 | 1.499 | 0.4048 | 2.4288 | 36 |
| 21 | | 7.1399 | 19.4928 | 91 |

$7.1399 = 6A + 21b$ , $19.4928 = 21A + 91b \Rightarrow A = 2.2893$, $b = -0.3141$, $a = e^A = 9.868$.

$$y = 9.87\,e^{-0.31\,x}$$

So the data follow the law $y \approx 10\,e^{-0.3x}$.

## 7. Quick Revision

**Formula sheet**

$u = \dfrac{x - x_0}{h}$ (forward, start of table), $v = \dfrac{x - x_n}{h}$ (backward, end of table).

Forward: $y_0 + u\Delta y_0 + \dfrac{u(u-1)}{2!}\Delta^2 y_0 + \cdots$ Backward: $y_n + v\nabla y_n + \dfrac{v(v+1)}{2!}\nabla^2 y_n + \cdots$

Lagrange for unequal spacing; sum of terms $y_i \prod \dfrac{x - x_j}{x_i - x_j}$.

Least squares: write the table of $\sum x, \sum y, \sum xy, \sum x^2$ FIRST, then the normal equations.

Exponential / power law: take logs, fit a straight line, then convert back.

- In exams always draw the difference table neatly and show $u$ (or $v$) before substituting.
- Interpolating polynomial is unique, so Newton and Lagrange give the same answer for the same points.

## 8. Practice Problems (with answers)

1. Given $f(1) = 1,\ f(2) = 8,\ f(3) = 27,\ f(4) = 64$ find $f(2.5)$ by Newton forward formula. Ans: 15.625
2. Find $f(2)$ from $f(0) = 1,\ f(1) = 3,\ f(3) = 13$ using Lagrange. Ans: 7 (polynomial $x^2 + x + 1$)
3. Fit a line to (0,1), (1,3), (2,2), (3,4). Ans: $y = 1.3 + 0.8x$
4. Show that $\Delta^2 y_0 = y_2 - 2y_1 + y_0$ and $E = 1 + \Delta$.
5. Fit $y = ab^x$ to the data x: 2,3,4,5,6 ; y: 144, 172.8, 207.4, 248.8,

298.5. *Hint: use* $\log y = \log a + x \log b$. *Ans:* $y \approx 100\,(1.2)^x$

# UNIT - III

## Numerical Solution of Ordinary Differential Equations

*(Initial Value Problems: Taylor, Picard, Euler, Modified Euler, Runge-Kutta)*

## 1. Introduction

*Many engineering problems lead to a first-order differential equation with an initial condition, called an INITIAL VALUE PROBLEM (IVP):*

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0$$

- *Analytical solutions exist only for special forms. When no closed form is available we compute $y$ at a set of points $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, ...NUMERICALLY ($h$ = step size).*
- *SINGLE-STEP methods (this unit) compute $y_{n+1}$ from $y_n$ only: Taylor, Picard, Euler, Modified Euler, Runge-Kutta.*
- *Accuracy improves when $h$ is smaller, or when the method is of higher order. A method has order $p$ if its local error is $O(h^{p+1})$.*

*Notation:* $x_n = x_0 + nh$, $y_n \approx y(x_n)$, $f_n = f(x_n, y_n)$.

## 2. Taylor's Series Method

*If $y(x)$ is analytic at $x_0$, expand it about $x_0$:*

## Taylor's series solution

$$y(x) = y_0 + (x - x_0)\,y_0' + \frac{(x-x_0)^2}{2!}\,y_0'' + \frac{(x-x_0)^3}{3!}\,y_0''' + \cdots$$

- *Obtain* $y_0' = f(x_0, y_0)$ *from the equation, and* $y''$, $y'''$, … *by DIFFERENTIATING the equation repeatedly, then put* $x = x_0,\ y = y_0$.
- *To go further, treat* $(x_1, y_1)$ *as a new starting point and expand again:* $y_{n+1} = y_n + hy_n' + \frac{h^2}{2!}y_n'' + \cdots$
- *Merits: accurate if many terms are taken. Demerit: derivatives become tedious.*

**Example 1.** $\frac{dy}{dx} = x^2 + y^2,\ y(0) = 1$ **: find** $y(0.1)$ **and** $y(0.2)$ **(up to** $x^4$ **term)**

$y' = x^2 + y^2 \ \Rightarrow y'(0) = 0 + 1 = 1$

$y'' = 2x + 2yy' \ \Rightarrow y''(0) = 0 + 2(1)(1) = 2$

$y''' = 2 + 2(y'^2 + yy'') \ \Rightarrow y'''(0) = 2 + 2(1+2) = 8$

$y^{iv} = 2(2y'y'' + y'y'' + yy''') = 6y'y'' + 2yy''' \ \Rightarrow$

$y^{iv}(0) = 12 + 16 = 28$

$$y(x) = 1 + x + \frac{2x^2}{2!} + \frac{8x^3}{3!} + \frac{28x^4}{4!} = 1 + x + x^2 + \frac{4}{3}x^3 + \frac{7}{6}x^4 + \cdots$$

$y(0.1) = 1 + 0.1 + 0.01 + 0.001333 + 0.000117 = 1.1115;$

$y(0.2) = 1 + 0.2 + 0.04 + 0.010667 + 0.001867 = 1.2525.$

**Example 2.** $y' = x + y,\ y(0) = 1$ **: find** $y(0.1)$ **by Taylor's method**

$y' = x + y \Rightarrow 1; \quad y'' = 1 + y' \Rightarrow 2; \quad y''' = y'' \Rightarrow 2; \quad y^{iv} = 2$ *(all at* $x = 0$*).*

$$y(x) = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \cdots$$

$y(0.1) = 1 + 0.1 + 0.01 + 0.000333 + 0.0000083 = 1.110342$ *(exact* $2e^{0.1} - 1.1 = 1.110342$*).*

## 3. Picard's Method of Successive Approximations

*Integrate* $\dfrac{dy}{dx} = f(x, y)$ *from* $x_0$ *to* $x$ *:* $y = y_0 + \displaystyle\int_{x_0}^{x} f(x, y)\, dx$*. The unknown* $y$ *on the right is replaced step by step:*

**Picard's iteration**

$$y^{(1)} = y_0 + \int_{x_0}^{x} f(x, y_0)\, dx, \qquad y^{(2)} = y_0 + \int_{x_0}^{x} f(x, y^{(1)})\, dx$$

$$y^{(n+1)} = y_0 + \int_{x_0}^{x} f(x, y^{(n)})\, dx, \qquad n = 0, 1, 2, \ \ldots$$

- *Start with* $y^{(0)} = y_0$ *(a constant). Stop when two successive approximations agree to the required accuracy in the interval.*
- *Merit: gives* $y$ *as a function of* $x$*. Demerit: integrals become difficult for complicated* $f$*.*

**Example 3.** $y' = x + y,\ y(0) = 1$ **: find** $y^{(1)}, y^{(2)}, y^{(3)}$ **and** $y(0.1)$

$$y^{(1)} = 1 + \int_0^x (x + 1)\, dx = 1 + x + \frac{x^2}{2}$$

$$y^{(2)} = 1 + \int_0^x \left(x + 1 + x + \frac{x^2}{2}\right) dx = 1 + x + x^2 + \frac{x^3}{6}$$

$$y^{(3)} = 1 + \int_0^x \left(1 + 2x + x^2 + \frac{x^3}{6}\right) dx = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}$$

$y(0.1) \approx 1.110337$ *(exact 1.110342).*

**Example 4.** $y' = x^2 + y^2,\ y(0) = 0$ **: find** $y(0.4)$ **using Picard**

$$y^{(1)} = \int_0^x x^2 dx = \frac{x^3}{3}\ ;\ y^{(2)} = \int_0^x \left(x^2 + \frac{x^6}{9}\right) dx = \frac{x^3}{3} + \frac{x^7}{63}$$

$$y^{(3)} = \frac{x^3}{3} + \frac{x^7}{63} + \frac{2x^{11}}{2079} + \frac{x^{15}}{59535}$$

$y(0.4) \approx 0.02136$ *(two terms:* $0.02133 + 0.00026$*). Four-decimal accuracy: 0.0213.*

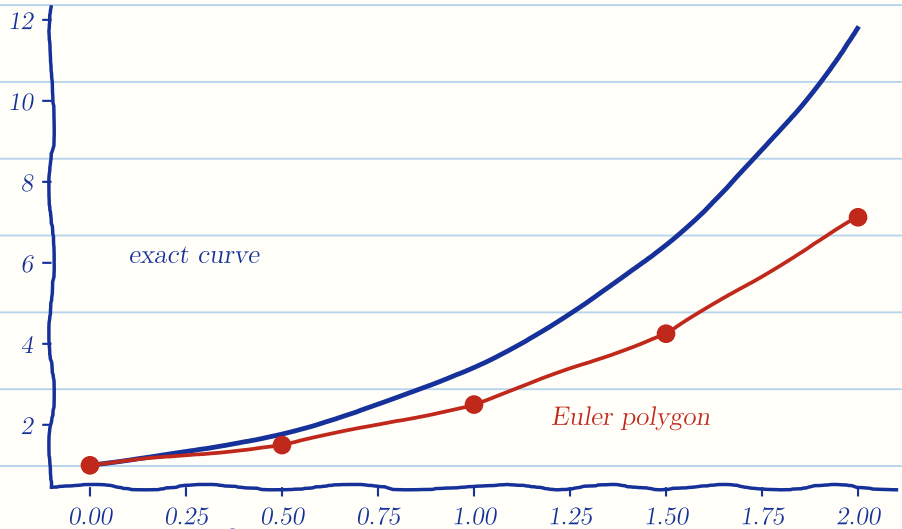
## 4. Euler's Method

*Simplest method. On* $[x_n, x_{n+1}]$ *replace the solution curve by its TANGENT at* $(x_n, y_n)$ *(slope* $f(x_n, y_n)$*):*

**Euler's formula**

$$y_{n+1} = y_n + h\, f(x_n, y_n), \qquad n = 0, 1, 2, \ \ldots$$

*Euler: tangent steps lag below the curve*




- *Local error* $O(h^2)$*, global error* $O(h)$ *- FIRST order. Needs small* $h$*.*
- *Geometrically the curve is replaced by a polygon; error accumulates and for convex solution curves Euler values fall below the true values.*

**Example 5.** $\frac{dy}{dx} = x + y,\ y(0) = 1$ **: find** $y(0.5)$ **taking** $h = 0.1$

| $n$ | $x_n$ | $y_n$ | $f = x_n + y_n$ | $y_{n+1} = y_n + 0.1f$ |
|---|---|---|---|---|
| 0 | 0.0 | 1.0000 | 1.0000 | 1.1000 |
| 1 | 0.1 | 1.1000 | 1.2000 | 1.2200 |
| 2 | 0.2 | 1.2200 | 1.4200 | 1.3620 |
| 3 | 0.3 | 1.3620 | 1.6620 | 1.5282 |
| 4 | 0.4 | 1.5282 | 1.9282 | 1.7210 |

$y(0.5) \approx 1.7210$ *by Euler; exact* $= 2e^{0.5} - 1.5 = 1.7974$. *Error* $\approx 0.0764$ *(large - so refine).*

**Example 5(b).** $\frac{dy}{dx} = \frac{y - x}{y + x},\ y(0) = 1$ **: find** $y(0.1)$ **taking** $h = 0.02$

| $n$ | $x_n$ | $y_n$ | $f(x_n, y_n)$ | $y_{n+1}$ |
|---|---|---|---|---|
| 0 | 0.00 | 1.00000 | 1.00000 | 1.02000 |
| 1 | 0.02 | 1.02000 | 0.96154 | 1.03923 |
| 2 | 0.04 | 1.03923 | 0.92587 | 1.05775 |
| 3 | 0.06 | 1.05775 | 0.89264 | 1.07560 |
| 4 | 0.08 | 1.07560 | 0.86154 | 1.09283 |

$y(0.1) \approx 1.0928$.

**Where does Euler's formula come from?**

*Taylor:* $y(x_n + h) = y_n + hy'_n + \frac{h^2}{2}y''_n + \cdots$. *Keeping only the first two terms gives Euler; the dropped term* $\frac{h^2}{2}y''(\xi)$ *is the LOCAL TRUNCATION ERROR. Halving* $h$ *halves the final error but doubles the number of steps.*

## 5. Modified Euler's Method (Heun / Predictor-Corrector)

*Euler uses the slope at the LEFT end only. Improve by using the AVERAGE of slopes at both ends (trapezoidal rule):*

**Modified Euler formulae**

$$\text{Predictor:} \quad y_{n+1}^{(0)} = y_n + h\, f(x_n, y_n)$$

$$\text{Corrector:} \quad y_{n+1}^{(k+1)} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f\left(x_{n+1}, y_{n+1}^{(k)}\right)\right]$$

- *Repeat the corrector for $k = 0, 1, 2, \ldots$ until two successive values of $y_{n+1}$ agree.*
- *Second-order method: local error $O(h^3)$, global error $O(h^2)$.*

**Example 6.** $y' = x + y,\ y(0) = 1$ **: find** $y(0.1)$ **and** $y(0.2)$ $(h = 0.1)$

| Value | Computation | Result |
|---|---|---|
| $y_1^{(0)}$ | *1.00000 + 0.1(1.00000)* | *1.10000* |
| $y_1^{(1)}$ | *avg slope with 1.10000* | *1.11000* |
| $y_1^{(2)}$ | *avg slope with 1.11000* | *1.11050* |
| $y_2^{(0)}$ | *1.11050 + 0.1(1.21050)* | *1.23155* |
| $y_2^{(1)}$ | *avg slope with 1.23155* | *1.24260* |
| $y_2^{(2)}$ | *avg slope with 1.24260* | *1.24316* |

$y(0.1) \approx 1.11000$, $y(0.2) \approx 1.24316$ *(exact 1.24281). Much better than Euler.*

**Example 7. Compare errors at $x = 0.5$: $y' = x + y$, $y(0) = 1$, $h = 0.1$**

| Method | Order | $y(0.5)$ | Error |
|---|---|---|---|
| *Euler* | *1* | *1.721020* | *0.076423* |
| *Modified Euler (1 corr.)* | *2* | *1.794894* | *0.002549* |
| *RK 4th order* | *4* | *1.797441* | *1.26e-06* |
| *Exact* | - | *1.797443* | *0* |

## 6. Runge-Kutta Methods

*Taylor-quality accuracy WITHOUT computing higher derivatives: use several slope evaluations $k_1, k_2, \ldots$ in each step.*

### (a) Second-order Runge-Kutta

**RK-2 (improved Euler)**

$$k_1 = h\, f(x_n, y_n), \qquad k_2 = h\, f(x_n + h,\ y_n + k_1)$$

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$$

*(Another RK-2 form uses the mid-point: $k_2 = hf(x_n + \frac{h}{2}, y_n + \frac{k_1}{2})$, $y_{n+1} = y_n + k_2$.) Local error $O(h^3)$.*

### (b) Fourth-order Runge-Kutta (the most widely used)

## RK-4 formulae

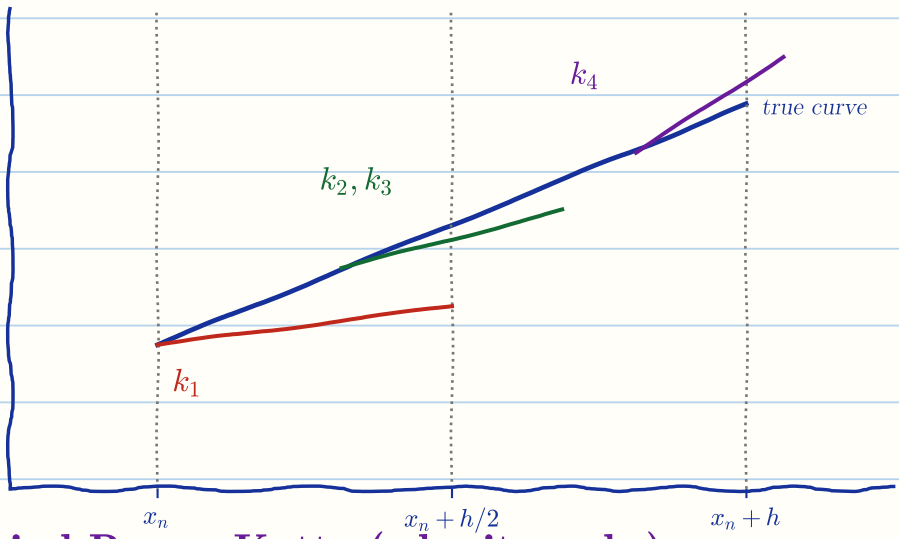
$$k_1 = hf(x_n, y_n), \qquad k_2 = hf\left(x_n + \frac{h}{2},\ y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2},\ y_n + \frac{k_2}{2}\right), \qquad k_4 = hf(x_n + h,\ y_n + k_3)$$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

- *Local error $O(h^5)$, global error $O(h^4)$. Needs 4 function values per step but NO derivatives of $f$.*
- *$k_1$: slope at start, $k_2, k_3$: slopes at the mid-point (two estimates), $k_4$: slope at the end. Weights 1:2:2:1.*

*RK-4: four slopes are sampled and averaged*




## Idea behind Runge-Kutta (why it works)

*Assume $y_{n+1} = y_n + ak_1 + bk_2$ with $k_1 = hf(x_n, y_n)$, $k_2 = hf(x_n + \alpha h,\ y_n + \beta k_1)$. Expand $k_2$ by Taylor and match with the Taylor series of $y$ up to $h^2$ terms: this forces*

$$a + b = 1, \qquad b\alpha = \frac{1}{2}, \qquad b\beta = \frac{1}{2}$$

*Choosing $a = b = \frac{1}{2}$, $\alpha = \beta = 1$ gives RK-2 above; choosing*

$a=0, b=1, \alpha=\beta=\frac{1}{2}$ *gives the mid-point form. RK-4 matches Taylor up to* $h^4$ *in the same way.*

**Example 8. RK-2:** $y'=x+y,\ y(0)=1$ **: find** $y(0.2)$ **taking** $h=0.2$

$k_1=0.2(0+1)=0.2000$ ; $k_2=0.2\,f(0.2,\ 1.2)=0.2(1.4)=0.2800$

$y(0.2)=1+\frac{1}{2}(0.20+0.28)=1.2400$ *(exact 1.2428).*

**Example 9. RK-4:** $y'=x+y,\ y(0)=1$ **: find** $y(0.2)$ **taking** $h=0.2$

$k_1=0.2(0+1)=0.2000$

$k_2=0.2\,f(0.1,\ 1+0.10)=0.2(0.1+1.1)=0.2400$

$k_3=0.2\,f(0.1,\ 1+0.12)=0.2(0.1+1.12)=0.2440$

$k_4=0.2\,f(0.2,\ 1+0.244)=0.2(0.2+1.244)=0.2888$

$y(0.2)=1+\frac{1}{6}(0.20+2(0.24)+2(0.244)+0.2888)=1.24280$ *(exact 1.24281).*

**Example 10. RK-4:** $y'=\dfrac{y^2-x^2}{y^2+x^2},\ y(0)=1$ **: find** $y(0.2)$ **with** $h=0.2$

| | Expression | Value |
|---|---|---|
| $k_1$ | $h\,f(0,1)$ | *0.20000* |
| $k_2$ | $h\,f(0.1,\ 1+k_1/2)$ | *0.19672* |
| $k_3$ | $h\,f(0.1,\ 1+k_2/2)$ | *0.19671* |
| $k_4$ | $h\,f(0.2,\ 1+k_3)$ | *0.18913* |

$y(0.2)=1+\frac{1}{6}(k_1+2k_2+2k_3+k_4)=1.19600.$

*(High-accuracy check: 1.19601.)*

**Example 11. RK-4 for two steps: $y' = x + y,\ y(0) = 1,\ h = 0.1$ : find $y(0.2)$**

| $x_n$ | $y_n$ | $k_1$ | $k_2$ | $k_3$ | $k_4$ | $y_{n+1}$ |
|---|---|---|---|---|---|---|
| 0.0 | 1.00000 | 0.10000 | 0.11000 | 0.11050 | 0.12105 | 1.11034 |
| 0.1 | 1.11034 | 0.12103 | 0.13209 | 0.13264 | 0.14430 | 1.24281 |

$y(0.2) = 1.24281$ *agrees with exact 1.24281 to 5 decimals.*

**Example 12. RK-4: $y' = 3x + \dfrac{y}{2},\ y(0) = 1$ : find $y(0.2)$ with $h = 0.2$**

| $k_1$ | $k_2$ | $k_3$ | $k_4$ | $y(0.2)$ |
|---|---|---|---|---|
| 0.10000 | 0.16500 | 0.16825 | 0.23683 | 1.16722 |

*Here* $k_2 = 0.2\, f(0.1,\ 1.05) = 0.2(0.3 + 0.525),\ k_3 = 0.2\, f(0.1,\ 1 + k_2/2),$ $k_4 = 0.2\, f(0.2,\ 1 + k_3).$

### Choosing the step size / checking accuracy

- *Compute $y(x)$ with step $h$ and again with $h/2$. If the two answers agree to the required decimals, accept; otherwise halve again.*
- *For RK-4 the error of the $h/2$ result is about $\frac{1}{16}$ of the error of the $h$ result (order 4); for Euler only $\frac{1}{2}$.*
- *If $f$ changes rapidly (stiff problems) a very small $h$ is needed or implicit methods are used (beyond this syllabus).*

## 7. Comparison of Methods

| Method | Order | Evaluations / step | Remarks |
|---|---|---|---|
| Taylor | depends on terms | 1 (+ derivatives) | needs $y', y'', \ldots$ |
| Picard | series in $x$ | integrals | gives $y(x)$ |
| Euler | 1 | 1 | simple, least accurate |
| Modified Euler | 2 | 2 (per iteration) | predictor-corrector |
| RK-2 | 2 | 2 | no derivatives |
| RK-4 | 4 | 4 | most popular, accurate |

**Formula sheet**

Taylor: $y = y_0 + hy_0' + \frac{h^2}{2!}y_0'' + \cdots$ Picard:

$y^{(n+1)} = y_0 + \int f(x, y^{(n)})dx$

Euler: $y_{n+1} = y_n + hf_n$; Modified Euler:

$y_{n+1} = y_n + \frac{h}{2}[f_n + f(x_{n+1}, y_{n+1}^{(0)})]$

RK-4: $y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

- In exams show a neat table of $k_1 \ldots k_4$; keep at least 4-5 decimal places; always state the final $y$ with units of accuracy asked.
- Typical questions: Taylor/Picard (5 marks), Euler or Modified Euler (5 marks), RK-4 (10 marks).

## Practice problems

1. Use Euler's method: $y' = 1 + y^2$, $y(0) = 0$, find $y(0.6)$ with $h = 0.2$. Ans: $\approx 0.6413$
2. Picard: $y' = x - y^2$, $y(0) = 1$, find $y^{(2)}(x)$ and $y(0.1)$. Ans: $1 - x + \frac{3x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} - \frac{x^5}{20}$ ; $y(0.1) \approx 0.9144$
3. Modified Euler: $y' = \log_{10}(x + y)$, $y(0) = 2$, $h = 0.2$, find $y(0.2)$ [Ans: 2.0656].

4. *RK-4:* $y' = x^2 + y^2,\ y(1) = 1.5,\ h = 0.1$*: find* $y(1.1)$ *[Ans: 1.8955].*

# UNIT - IV

## Laplace Transforms

(Definition, standard transforms, shifting theorems, derivatives & integrals, unit step, convolution, periodic functions)

## 1. Definition & Existence

Let $f(t)$ be defined for $t \geq 0$. The LAPLACE TRANSFORM of $f(t)$ is

**Definition**

$$L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

- $s$ is a real (or complex) parameter; the integral must converge.
  $f(t)$ is the INVERSE transform: $f(t) = L^{-1}\{F(s)\}$.
- Laplace transform converts a differential equation in $t$ into an ALGEBRAIC equation in $s$ - very useful for circuits, vibrations, control.

### Existence theorem (sufficient conditions)

- $f(t)$ is PIECEWISE CONTINUOUS on every finite interval $[0, b]$ (finite jumps only).
- $f(t)$ is of EXPONENTIAL ORDER $a$: $|f(t)| \leq Me^{at}$ for all $t > T$. Then $L\{f\}$ exists for $s > a$.

These are sufficient, not necessary: e.g. $f = t^{-1/2}$ is not bounded at 0 but its transform exists. Functions like $e^{t^2}$ or $\tan t$ have NO Laplace transform.

### Linearity

$$L\{af(t)+bg(t)\}=aL\{f(t)\}+bL\{g(t)\}$$

## Derivation of basic transforms

(i) $L\{1\}=\int_0^\infty e^{-st}dt=\left[\frac{e^{-st}}{-s}\right]_0^\infty=\frac{1}{s},\ s>0.$

(ii) $L\{e^{at}\}=\int_0^\infty e^{-(s-a)t}dt=\frac{1}{s-a},\ s>a.$

(iii) $L\{t^n\}=\int_0^\infty e^{-st}t^n dt=\frac{\Gamma(n+1)}{s^{n+1}}=\frac{n!}{s^{n+1}}$ (put $st=x$).

(iv) $L\{\sin at\}$: $L\{e^{iat}\}=\frac{1}{s-ia}=\frac{s+ia}{s^2+a^2}$; equate real and imaginary parts $\Rightarrow L\{\cos at\}=\frac{s}{s^2+a^2}, L\{\sin at\}=\frac{a}{s^2+a^2}.$

## 2. Laplace Transforms of Standard Functions

| $f(t)$ | $F(s)=L\{f\}$ | $f(t)$ | $F(s)$ |
|---|---|---|---|
| $1$ | $1/s$ | $\sinh at$ | $a/(s^2-a^2)$ |
| $t^n$ ($n$ integer) | $n!/s^{n+1}$ | $\cosh at$ | $s/(s^2-a^2)$ |
| $e^{at}$ | $1/(s-a)$ | $te^{at}$ | $1/(s-a)^2$ |
| $\sin at$ | $a/(s^2+a^2)$ | $e^{at}\sin bt$ | $b/[(s-a)^2+b^2]$ |
| $\cos at$ | $s/(s^2+a^2)$ | $e^{at}\cos bt$ | $(s-a)/[(s-a)^2+b^2]$ |
| $t\sin at$ | $2as/(s^2+a^2)^2$ | $t\cos at$ | $(s^2-a^2)/(s^2+a^2)^2$ |
| $t^{-1/2}$ | $\sqrt{\pi/s}$ | $t^{1/2}$ | $\sqrt{\pi}/(2s^{3/2})$ |

## Example 1. Find the Laplace transforms

1. $L\{3t^2-4t+5+2e^{-3t}\}=\frac{6}{s^3}-\frac{4}{s^2}+\frac{5}{s}+\frac{2}{s+3}$

2. $\sin 2t\cos 3t=\frac{1}{2}(\sin 5t-\sin t)\ \Rightarrow L=\frac{1}{2}\left[\frac{5}{s^2+25}-\frac{1}{s^2+1}\right]$

3. $\cos^2 t=\frac{1}{2}(1+\cos 2t)\ \Rightarrow L=\frac{1}{2}\left[\frac{1}{s}+\frac{s}{s^2+4}\right]=\frac{s^2+2}{s(s^2+4)}$

4. $\sin^3 t=\frac{1}{4}(3\sin t-\sin 3t)\ \Rightarrow$

$$L=\frac{1}{4}\left[\frac{3}{s^2+1}-\frac{3}{s^2+9}\right]=\frac{6}{(s^2+1)(s^2+9)}$$

5. $\cosh^2 2t = \frac{1}{2}(1+\cosh 4t) \Rightarrow L = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2-16}\right]$

## 3. Properties of Laplace Transforms

### (a) First Shifting Theorem (shift in $s$)

**First shifting theorem**

*If* $L\{f(t)\} = F(s)$ *then*

$$L\{e^{at} f(t)\} = F(s-a)$$

*i.e. replace* $s$ *by* $s-a$ *in* $F(s)$ *(and by* $s+a$ *for* $e^{-at}$*).*

*Proof:* $L\{e^{at} f\} = \int_0^\infty e^{-(s-a)t} f(t)dt = F(s-a).$

**Example 2.**

1. $L\{e^{2t}\cos 3t\} = \dfrac{s-2}{(s-2)^2+9}$
2. $L\{e^{-t}t^3\} = \dfrac{3!}{(s+1)^4} = \dfrac{6}{(s+1)^4}$
3. $L\{e^{-3t}\sin 2t\} = \dfrac{2}{(s+3)^2+4}$
4. $L\{e^t(3\cosh 2t - 5\sinh 2t)\} = \dfrac{3(s-1)-10}{(s-1)^2-4} = \dfrac{3s-13}{s^2-2s-3}$

### (b) Change of scale

$$L\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right), \quad a > 0$$

### (c) Multiplication by $t^n$ (derivative of the transform)

$$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

**Example 3.**

1. $L\{t\sin 2t\} = -\dfrac{d}{ds}\left(\dfrac{2}{s^2+4}\right) = \dfrac{4s}{(s^2+4)^2}$
2. $L\{t^2\cos t\} = \dfrac{d^2}{ds^2}\left(\dfrac{s}{s^2+1}\right) = \dfrac{2s^3-6s}{(s^2+1)^3}$
3. $L\{te^{-t}\sin t\}$: *first* $L\{t\sin t\} = \dfrac{2s}{(s^2+1)^2}$*; shift* $s \to s+1$*:*

$$\frac{2(s+1)}{[(s+1)^2+1]^2}$$

### (d) Division by $t$

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du$$

1. $L\left\{\dfrac{\sin t}{t}\right\} = \displaystyle\int_s^\infty \frac{du}{u^2+1} = \tan^{-1}\frac{1}{s} = \cot^{-1} s$
2. $L\left\{\dfrac{1-\cos t}{t}\right\} = \displaystyle\int_s^\infty \left(\frac{1}{u} - \frac{u}{u^2+1}\right)du = \frac{1}{2}\ln\frac{s^2+1}{s^2}$
3. $L\left\{\dfrac{e^{-at}-e^{-bt}}{t}\right\} = \ln\dfrac{s+b}{s+a}$

### Properties at a glance

| Operation on $f(t)$ | Effect on $F(s)$ |
|---|---|
| *multiply by* $e^{at}$ | *shift:* $F(s-a)$ |
| *multiply by* $t^n$ | $(-1)^n F^{(n)}(s)$ |
| *divide by* $t$ | $\displaystyle\int_s^\infty F(u)du$ |
| *differentiate* | $sF(s) - f(0)$ |
| *integrate* $0 \to t$ | $F(s)/s$ |
| *delay by* $a$: $f(t-a)u(t-a)$ | $e^{-as}F(s)$ |
| *convolution* $f * g$ | $F(s)\,G(s)$ |

## 4. Transforms of Derivatives and Integrals

### Derivatives

$$L\{f'(t)\} = sF(s) - f(0)$$

$$L\{f''(t)\} = s^2F(s) - s\,f(0) - f'(0)$$

$$L\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

*Proof for $f'$ (integration by parts):*

$$\int_0^\infty e^{-st}f'dt = [e^{-st}f]_0^\infty + s\int_0^\infty e^{-st}f\,dt = -f(0) + sF(s).$$

**Integrals**

$$L\left(\int_0^t f(u)\,du\right)=\frac{F(s)}{s}$$

**Example 4.**

1. *Find* $L\{\sin^2 t\}$ *using derivatives. Let* $f=\sin^2 t$, $f(0)=0$, $f'=\sin 2t$. *Then* $L\{f'\}=sF$ *gives* $\dfrac{2}{s^2+4}=sF\Rightarrow F=\dfrac{2}{s(s^2+4)}$.
2. $L\left(\displaystyle\int_0^t \frac{\sin u}{u}\,du\right)=\frac{1}{s}\tan^{-1}\frac{1}{s}$
3. $L\left(\displaystyle\int_0^t e^{-u}\cos u\,du\right)=\frac{1}{s}\cdot\frac{s+1}{(s+1)^2+1}$

## 5. Inverse Laplace Transform

*If* $L\{f(t)\}=F(s)$ *then* $f(t)=L^{-1}\{F(s)\}$. *Inverse is also linear. Basic list (read the table of Section 2 backwards):*

| $F(s)$ | $L^{-1}\{F\}$ | $F(s)$ | $L^{-1}\{F\}$ |
|---|---|---|---|
| $1/s$ | $1$ | $1/(s^2+a^2)$ | $\frac{1}{a}\sin at$ |
| $1/s^n$ | $t^{n-1}/(n-1)!$ | $s/(s^2+a^2)$ | $\cos at$ |
| $1/(s-a)$ | $e^{at}$ | $1/(s^2-a^2)$ | $\frac{1}{a}\sinh at$ |
| $s/(s^2-a^2)$ | $\cosh at$ | $s/(s^2+a^2)^2$ | $\frac{t\sin at}{2a}$ |

*Methods: (1) partial fractions, (2) completing the square + first shift, (3) first shifting theorem, (4) convolution theorem, (5) derivative/integral rules.*

**Example 5. Find the inverse Laplace transforms**

1. $\dfrac{3s+7}{s^2-2s-3}=\dfrac{3s+7}{(s-3)(s+1)}=\dfrac{4}{s-3}-\dfrac{1}{s+1}\Rightarrow f(t)=4e^{3t}-e^{-t}$
2. $\dfrac{s+2}{s^2+4s+13}=\dfrac{s+2}{(s+2)^2+3^2}\Rightarrow f(t)=e^{-2t}\cos 3t$
3. $\dfrac{s}{(s^2+1)(s^2+4)}=\dfrac{1}{3}\left[\dfrac{s}{s^2+1}-\dfrac{s}{s^2+4}\right]\Rightarrow f(t)=\dfrac{1}{3}(\cos t-\cos 2t)$

4. $\frac{1}{s^2(s^2+1)} = \frac{1}{s^2} - \frac{1}{s^2+1} \Rightarrow f(t) = t - \sin t$
5. $\frac{2s+3}{s^2+6s+25} = \frac{2(s+3)-3}{(s+3)^2+16} \Rightarrow f(t) = e^{-3t}\left(2\cos 4t - \frac{3}{4}\sin 4t\right)$
6. $\ln\frac{s+1}{s-1}$: $F'(s) = \frac{1}{s+1} - \frac{1}{s-1} \Rightarrow L^{-1}\{F'\} = e^{-t} - e^{t} = -tf(t) \Rightarrow$
   $f(t) = \frac{2\sinh t}{t}$
7. $\frac{1}{s(s+1)^2} = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} \Rightarrow f(t) = 1 - e^{-t} - te^{-t}$
8. $\frac{5s+3}{(s-1)(s^2+2s+5)} = \frac{1}{s-1} - \frac{(s+1)-3}{(s+1)^2+4} \Rightarrow$
   $f(t) = e^{t} - e^{-t}\cos 2t + \frac{3}{2}e^{-t}\sin 2t$
9. $\frac{s^2}{(s^2+1)(s^2+4)} = \frac{1}{3}\left[\frac{4}{s^2+4} - \frac{1}{s^2+1}\right] \Rightarrow f(t) = \frac{1}{3}(2\sin 2t - \sin t)$

## Evaluation of integrals using Laplace transforms

*Put the required value of $s$ in* $\int_0^\infty e^{-st} f(t)dt = F(s)$.

1. $\int_0^\infty te^{-3t}\sin t\, dt = L\{t\sin t\}_{s=3} = \frac{2s}{(s^2+1)^2}|_{s=3} = \frac{6}{100} = 0.06$
2. $\int_0^\infty \frac{e^{-t} - e^{-2t}}{t} dt = \ln\frac{s+2}{s+1}|_{s=0} = \ln 2$
3. $\int_0^\infty \frac{\sin t}{t} dt = \cot^{-1} s|_{s=0} = \frac{\pi}{2}$
4. $\int_0^\infty e^{-2t}\cos 3t\, dt = \frac{s}{s^2+9}|_{s=2} = \frac{2}{13}$

## Initial and final value theorems (useful facts)

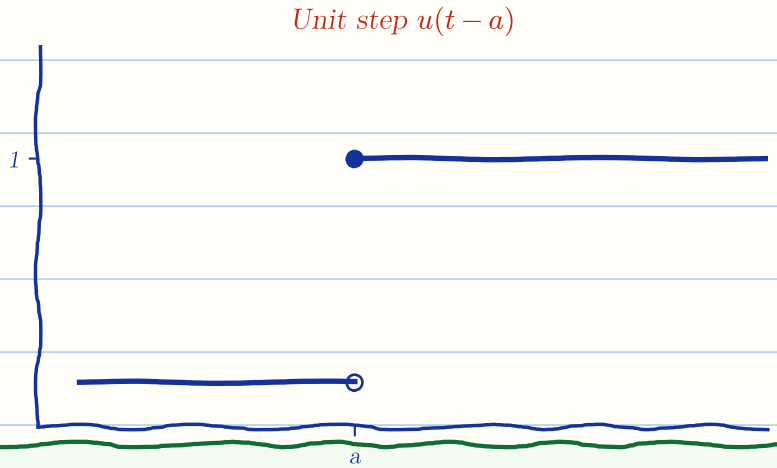
$$f(0^+) = \lim_{s\to\infty} sF(s), \qquad f(\infty) = \lim_{s\to 0} sF(s)$$

*Example:* $F(s) = \frac{3}{s(s+2)}$: $f(0) = \lim 3/(s+2) = 0$ ; $f(\infty) = 3/2$.

*(Check:* $f = \frac{3}{2}(1 - e^{-2t})$.*)*

# 6. Unit Step (Heaviside) Function & Second Shifting Theorem

*The UNIT STEP FUNCTION at $t=a$: $u(t-a)=0$ for $t<a$ and $u(t-a)=1$ for $t\geq a$.*




**Second shifting theorem**

$$L\{u(t-a)\}=\frac{e^{-as}}{s}$$

$$L\{f(t-a)\,u(t-a)\}=e^{-as}F(s) \quad \text{(Second shifting theorem)}$$

$$L^{-1}\{e^{-as}F(s)\}=f(t-a)\,u(t-a)$$

*Alternative form: $L\{g(t)\,u(t-a)\}=e^{-as}L\{g(t+a)\}$ (shift the function first).*

**Example 6.**

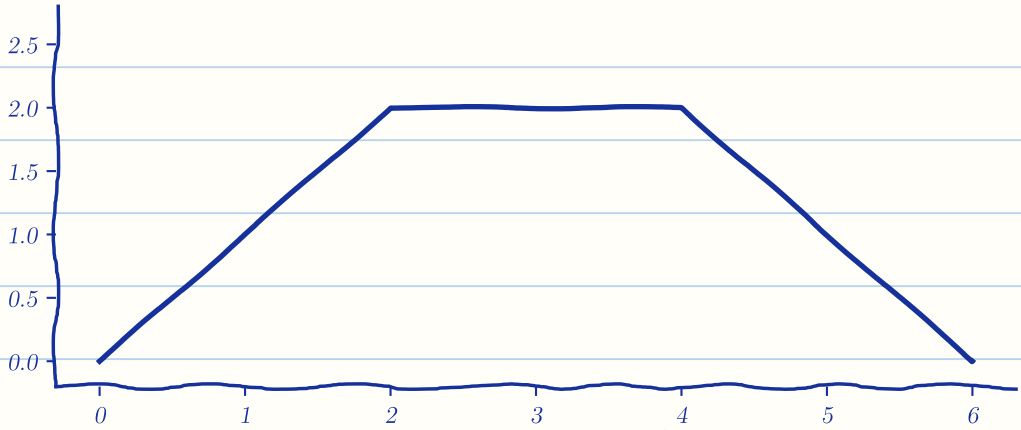
1. $L\{(t-2)^3u(t-2)\}=e^{-2s}\frac{3!}{s^4}=\frac{6e^{-2s}}{s^4}$
2. $f(t)=t^2$ *for* $t>2$, $0$ *otherwise.* $f=t^2u(t-2)$;
   $g(t+2)=(t+2)^2=t^2+4t+4 \Rightarrow L\{f\}=e^{-2s}\left[\frac{2}{s^3}+\frac{4}{s^2}+\frac{4}{s}\right]$
3. $f(t)=\sin t$ *for* $0<t<\pi$, $0$ *for* $t>\pi$.
   $f=\sin t-\sin t\,u(t-\pi)=\sin t+\sin(t-\pi)u(t-\pi) \Rightarrow$
   $L\{f\}=\frac{1+e^{-\pi s}}{s^2+1}$

4. $L^{-1}\left\{\dfrac{e^{-3s}}{s^2+4}\right\}=\dfrac{1}{2}\sin 2(t-3)\,u(t-3)$

5. $L^{-1}\left\{\dfrac{e^{-\pi s}}{s(s^2+1)}\right\}: L^{-1}\left\{\frac{1}{s(s^2+1)}\right\}=1-\cos t \;\Rightarrow$

$[1-\cos(t-\pi)]\,u(t-\pi)=(1+\cos t)\,u(t-\pi)$

$f(t)=t[u(t)-u(t-2)]+2[u(t-2)-u(t-4)]+(6-t)u(t-4)$




*A piecewise function is written with step functions: each new piece is switched ON by $u(t-a)$, and the old one switched OFF.*

## 7. Convolution Theorem

*The CONVOLUTION of $f$ and $g$ is* $(f*g)(t)=\displaystyle\int_0^t f(u)\,g(t-u)\,du$,

*with* $f*g=g*f$.

**Convolution theorem**

$$L\{f*g\}=L\{f\}\cdot L\{g\}=F(s)\,G(s)$$

$$L^{-1}\{F(s)\,G(s)\}=\int_0^t f(u)\,g(t-u)\,du$$

**Example 7. Use convolution to find**

1. $L^{-1}\left\{\dfrac{1}{s(s^2+1)}\right\}=1*\sin t=\displaystyle\int_0^t \sin u\,du=1-\cos t$

2. $L^{-1}\left\{\dfrac{1}{(s+1)(s-2)}\right\}=e^{-t}*e^{2t}=\displaystyle\int_0^t e^{-u}e^{2(t-u)}du=\frac{e^{2t}-e^{-t}}{3}$

3. $L^{-1}\left\{\dfrac{1}{s^2(s^2+1)}\right\}=t*\sin t=\displaystyle\int_0^t u\sin(t-u)\,du=t-\sin t$

4. $L^{-1}\left\{\dfrac{s}{(s^2+a^2)^2}\right\} = \cos at * \dfrac{\sin at}{a} = \dfrac{t\sin at}{2a}$

5. $L^{-1}\left\{\dfrac{1}{(s^2+a^2)^2}\right\} = \dfrac{\sin at}{a} * \dfrac{\sin at}{a} = \dfrac{\sin at - at\cos at}{2a^3}$

6. $L^{-1}\left\{\dfrac{s^2}{(s^2+4)^2}\right\} = \cos 2t * \cos 2t \ \ldots \ = \dfrac{1}{4}\sin 2t + \dfrac{t}{2}\cos 2t$

## 8. Laplace Transform of Periodic Functions

*If $f(t+T) = f(t)$ for all $t$ (period $T$), then*

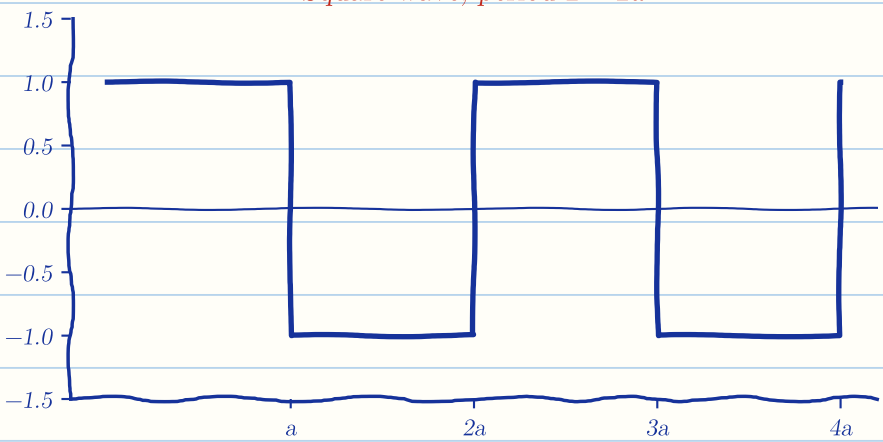
**Periodic function formula**

$$L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)\, dt$$

*Proof: split* $\int_0^\infty$ *into periods, put* $t = u + nT$, *sum the geometric series*

$$\sum e^{-nsT} = \frac{1}{1-e^{-sT}}.$$

*Square wave, period $T = 2a$*




### Example 8.

1. *SQUARE WAVE:* $f = 1$ $(0 < t < a)$, $f = -1$ $(a < t < 2a)$, $T = 2a$.

$$\int_0^{2a} e^{-st} f\, dt = \frac{(1-e^{-as})^2}{s}$$

$$L\{f\} = \frac{(1-e^{-as})^2}{s(1-e^{-2as})} = \frac{1-e^{-as}}{s(1+e^{-as})} = \frac{1}{s}\tanh\frac{as}{2}$$

2. *SAWTOOTH:* $f = t$ $(0 < t < a)$, *period* $a$.

$$\int_0^a te^{-st}\, dt = \frac{1}{s^2} - \frac{ae^{-as}}{s} - \frac{e^{-as}}{s^2}$$

$$L\{f\} = \frac{1}{s^2} - \frac{a\, e^{-as}}{s(1-e^{-as})}$$

3. *FULL-WAVE RECTIFIED SINE* $f=|\sin t|$, *period* $\pi$:

$$\int_0^{\pi} e^{-st}\sin t\,dt=\frac{1+e^{-\pi s}}{s^2+1}$$

$$L\{|\sin t|\}=\frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})}=\frac{1}{s^2+1}\coth\frac{\pi s}{2}$$

4. *HALF-WAVE RECTIFIED sine* (\$\sin t\$ *for* $0<t<\pi$, $0$ *for* $\pi<t<2\pi$, $T=2\pi$): $L\{f\}=\dfrac{1}{(s^2+1)(1-e^{-\pi s})}$

5. *TRIANGULAR WAVE:* $f=t$ $(0<t<a)$, $f=2a-t$ $(a<t<2a)$, $T=2a$: $L\{f\}=\dfrac{1}{s^2}\tanh\dfrac{as}{2}$

(*Since* $\displaystyle\int_0^{2a} e^{-st}f\,dt=\frac{(1-e^{-as})^2}{s^2}$ *and* $1-e^{-2as}=(1-e^{-as})(1+e^{-as})$.)

## 9. Solving Differential Equations (application)

*Procedure: (1) take* $L$ *of both sides using derivative formulae and initial conditions; (2) solve algebraically for* $Y(s)$; *(3) take* $L^{-1}$ *(partial fractions).*

**Example 9.** $y''+4y'+3y=e^{-t}$, $y(0)=0$, $y'(0)=0$

$$[s^2Y]+4[sY]+3Y=\frac{1}{s+1}\ \Rightarrow$$

$$Y=\frac{1}{(s+1)^2(s+3)}=\frac{-1/4}{s+1}+\frac{1/2}{(s+1)^2}+\frac{1/4}{s+3}$$

$$y(t)=-\frac{1}{4}e^{-t}+\frac{1}{2}te^{-t}+\frac{1}{4}e^{-3t}$$

*Check:* $y(0)=-\frac{1}{4}+\frac{1}{4}=0$, $y'(0)=\frac{1}{4}+\frac{1}{2}-\frac{3}{4}=0$. *OK.*

**Example 10.** $y''+y=\sin 2t$, $y(0)=y'(0)=0$

$$(s^2+1)Y=\frac{2}{s^2+4}\ \Rightarrow Y=\frac{2}{(s^2+1)(s^2+4)}=\frac{2}{3}\left[\frac{1}{s^2+1}-\frac{1}{s^2+4}\right]\Rightarrow$$

$$y=\frac{2}{3}\sin t-\frac{1}{3}\sin 2t$$

**Example 11.** $y' + 2y = u(t-1),\ y(0) = 0$

$(s+2)Y = \frac{e^{-s}}{s} \Rightarrow Y = e^{-s} \cdot \frac{1}{2}\left[\frac{1}{s} - \frac{1}{s+2}\right] \Rightarrow$

$y(t) = \frac{1}{2}(1 - e^{-2(t-1)})u(t-1)$

**Example 12.** $y'' + 9y = \cos 2t,\ y(0) = 1,\ y'(0) = 0$

$(s^2Y - s) + 9Y = \frac{s}{s^2+4} \Rightarrow$

$Y = \frac{s}{s^2+9} + \frac{s}{(s^2+4)(s^2+9)} = \frac{s}{s^2+9} + \frac{1}{5}\left[\frac{s}{s^2+4} - \frac{s}{s^2+9}\right]$

$$y(t) = \frac{4}{5}\cos 3t + \frac{1}{5}\cos 2t$$

**Example 13.** $y'' + 4y = u(t-2),\ y(0) = y'(0) = 0$ **(switch-on force)**

$Y = \frac{e^{-2s}}{s(s^2+4)}$, and $\frac{1}{s(s^2+4)} = \frac{1}{4}\left[\frac{1}{s} - \frac{s}{s^2+4}\right] \Rightarrow$

$y(t) = \frac{1}{4}[1 - \cos 2(t-2)]u(t-2)$

**Impulse (Dirac delta) - extra**

$\delta(t-a)$ *is a unit impulse at* $t = a$ *with* $\int f(t)\delta(t-a)dt = f(a)$ *and* $L\{\delta(t-a)\} = e^{-as}$. *Example:* $y'' + y = \delta(t-\pi),\ y(0) = y'(0) = 0 \Rightarrow$

$Y = \frac{e^{-\pi s}}{s^2+1} \Rightarrow y = \sin(t-\pi)\,u(t-\pi) = -\sin t\,u(t-\pi).$

## 10. Quick Revision Sheet

**Formulae**

$L\{t^n\} = \frac{n!}{s^{n+1}},\ L\{e^{at}\} = \frac{1}{s-a},\ L\{\sin at\} = \frac{a}{s^2+a^2},$
$L\{\cos at\} = \frac{s}{s^2+a^2}$
$L\{e^{at}f\} = F(s-a);\ L\{t^n f\} = (-1)^n F^{(n)}(s)\ ;\ L\{f/t\} = \int_s^\infty F\,du$
$L\{f'\} = sF - f(0)\ ;\ L\{f''\} = s^2F - sf(0) - f'(0)\ ;\ L\{\int f\} = F/s$
$L\{f(t-a)u(t-a)\} = e^{-as}F(s)\ ;\ L\{f*g\} = FG\ ;$ *periodic:* $\frac{\int_0^T e^{-st} f\,dt}{1-e^{-sT}}$

## Practice problems (with answers)

1. *Find* $L\{e^{-2t}\sin 3t\cos t\}$. *Ans:* $\frac{2}{(s+2)^2+16} + \frac{1}{(s+2)^2+4}$
2. *Find* $L\{t^2e^{-t}\sin t\}$ *(hint:* $\frac{d^2}{ds^2}$ *of* $L\{e^{-t}\sin t\}$*). Ans:* $\frac{6(s+1)^2-2}{[(s+1)^2+1]^3}$
3. *Find* $L^{-1}\left\{\frac{s+1}{s^2+s+1}\right\}$. *Ans:* $e^{-t/2}\left[\cos\frac{\sqrt{3}}{2}t + \frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right]$
4. *Solve* $y'' - 3y' + 2y = 4e^{2t},\ y(0) = -3,\ y'(0) = 5$. *Ans:* $y = 4(t+1)e^{2t} - 7e^t$
5. *Find* $L^{-1}\left\{\frac{1}{(s-1)(s^2+4)}\right\}$ *by convolution. Ans:* $\frac{1}{5}\left(e^t - \cos 2t - \frac{1}{2}\sin 2t\right)$

## Common mistakes to avoid

- *Forgetting initial conditions in* $L\{y'\}$, $L\{y''\}$. *Using* $s+a$ *instead of* $s-a$ *in the first shift for* $e^{at}$.
- *Writing* $L\{fg\} = L\{f\}L\{g\}$: *WRONG - product in* $t$ *is NOT product in* $s$ *(only convolution is).*
- *In the second shift the function must be written in terms of* $(t-a)$ *before taking the transform.*
- *For inverse problems: (i) factor denominator, (ii) partial fractions, (iii) complete the square, (iv) shift theorems.*

# UNIT - V

## Fourier Series & Fourier Transforms

*(Euler coefficients, even/odd, arbitrary interval, half-range, Fourier integral, sine/cosine transforms)*

## PART A : FOURIER SERIES

### 1. Introduction & Dirichlet's Conditions

*A function $f(x)$ is PERIODIC with period $T$ if $f(x+T)=f(x)$. A periodic function of period $2\pi$ can be represented as a series of sines and cosines:*

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)$$

*This is the FOURIER SERIES of $f(x)$; $a_0, a_n, b_n$ are FOURIER COEFFICIENTS.*

**Dirichlet's conditions (sufficient for the series to exist and converge)**

1. *$f(x)$ is single-valued, finite and periodic in the interval $(c, c+2\pi)$.*
2. *$f(x)$ has a finite number of discontinuities (finite jumps) in one period.*
3. *$f(x)$ has a finite number of maxima and minima in one period.*

**Convergence**

*Then the Fourier series converges: to $f(x)$ at every point of CONTINUITY, and*

$$\text{to } \frac{1}{2}[f(x^-)+f(x^+)] \text{ at a point of DISCONTINUITY}$$

*At end points $x=c$ and $x=c+2\pi$ the sum is $\frac{1}{2}[f(c)+f(c+2\pi)]$.*

## 2. Euler's Formulae for the Coefficients

*Orthogonality over a period of length $2\pi$ gives (for $m, n$ positive integers):*

$$\int_c^{c+2\pi} \sin nx\,dx = \int_c^{c+2\pi} \cos nx\,dx = 0, \qquad \int \sin mx \cos nx\,dx = 0$$

$$\int_c^{c+2\pi} \sin mx \sin nx\,dx = \int_c^{c+2\pi} \cos mx \cos nx\,dx = 0 \text{ for } m \neq n \text{ and } = \pi$$

*for $m = n$.*

**Euler's formulae (interval c to c+2pi)**

$$a_0 = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\cos nx\,dx, \qquad b_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\sin nx\,dx$$

- *Most common intervals: $(0, 2\pi)$ with $c = 0$ and $(-\pi, \pi)$ with $c = -\pi$.*
- *Multiply $f(x) = \frac{a_0}{2} + \sum \ldots$ by $\cos nx$ (or $\sin nx$), integrate over a period; only one term survives - that gives the formulae.*

### Useful tools for integration

*Bernoulli's generalised integration by parts:*

$\int u\,v\,dx = uv_1 - u'v_2 + u''v_3 - \cdots$ *where $v_1, v_2, \ldots$ are successive integrals of $v$ and $u', u'', \ldots$ successive derivatives of $u$ (stops when $u^{(k)} = 0$).*

| $n$ **integer** | $\sin n\pi$ | $\cos n\pi$ | $\sin 2n\pi$ | $\cos 2n\pi$ | $\sin \frac{n\pi}{2}$ |
|---|---|---|---|---|---|
| value | 0 | $(-1)^n$ | 0 | 1 | $0, 1, 0, -1, \ldots$ |

*Also* $\int e^{ax} \sin bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\sin bx - b\cos bx)$,

$\int e^{ax} \cos bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\cos bx + b\sin bx)$.

**Example 1. Expand $f(x)=x^2$ in the interval $(0,2\pi)$**

$$a_0=\frac{1}{\pi}\int_0^{2\pi}x^2dx=\frac{8\pi^2}{3},\ a_n=\frac{1}{\pi}\int_0^{2\pi}x^2\cos nx\,dx=\frac{4}{n^2},$$
$$b_n=\frac{1}{\pi}\int_0^{2\pi}x^2\sin nx\,dx=-\frac{4\pi}{n}$$

$$x^2=\frac{4\pi^2}{3}+\sum_{n=1}^{\infty}\left[\frac{4}{n^2}\cos nx-\frac{4\pi}{n}\sin nx\right],\quad 0<x<2\pi$$

*At $x=0$ (end point) the sum is $\frac{1}{2}[f(0)+f(2\pi)]=2\pi^2$ so*

$$2\pi^2=\tfrac{4\pi^2}{3}+4\textstyle\sum\frac{1}{n^2}\ \Rightarrow\ \sum\frac{1}{n^2}=\frac{\pi^2}{6}.$$

**Example 2. Expand $f(x)=e^{bx}$ in $(0,2\pi)$**

*With $K=e^{2\pi b}-1$:* $a_0=\frac{K}{\pi b},\ a_n=\frac{b\,K}{\pi(b^2+n^2)},\ b_n=-\frac{n\,K}{\pi(b^2+n^2)}$

$$e^{bx}=\frac{K}{\pi}\left[\frac{1}{2b}+\sum_{n=1}^{\infty}\frac{b\cos nx-n\sin nx}{b^2+n^2}\right]$$

**Example 3. $f(x)=-\pi$ for $-\pi<x<0$ ; $f(x)=x$ for $0<x<\pi$**

$$a_0=\frac{1}{\pi}\left[\int_{-\pi}^{0}(-\pi)dx+\int_0^{\pi}x\,dx\right]=\frac{1}{\pi}\left[-\pi^2+\frac{\pi^2}{2}\right]=-\frac{\pi}{2}$$
$$a_n=\frac{1}{\pi}\left[0+\int_0^{\pi}x\cos nx\,dx\right]=\frac{(-1)^n-1}{\pi n^2},$$
$$b_n=\frac{1}{\pi}\left[-\pi\frac{(-1)^n-1}{n}-\frac{\pi(-1)^n}{n}\right]=\frac{1-2(-1)^n}{n}$$

$$f(x)=-\frac{\pi}{4}+\sum_{n=1}^{\infty}\left[\frac{(-1)^n-1}{\pi n^2}\cos nx+\frac{1-2(-1)^n}{n}\sin nx\right]$$

*At $x=0$ the sum $=\frac{1}{2}[f(0^-)+f(0^+)]=\frac{1}{2}(-\pi+0)=-\frac{\pi}{2}$ (point of discontinuity).*

## 3. Even and Odd Functions

*$f$ is EVEN if $f(-x)=f(x)$ (symmetric about the y-axis), ODD if $f(-x)=-f(x)$. For the interval $(-\pi,\pi)$:*

| | Even function | Odd function |
|---|---|---|
| $a_0$ | $\frac{2}{\pi}\int_0^{\pi} f\,dx$ | $0$ |
| $a_n$ | $\frac{2}{\pi}\int_0^{\pi} f\cos nx\,dx$ | $0$ |
| $b_n$ | $0$ | $\frac{2}{\pi}\int_0^{\pi} f\sin nx\,dx$ |
| *Series* | *COSINE series only* | *SINE series only* |

- *Products:* even×even = even, odd×odd = even, even×odd = odd.

$\int_{-a}^{a}$ *of an odd function* $= 0$; *of an even function* $= 2\int_0^a$.

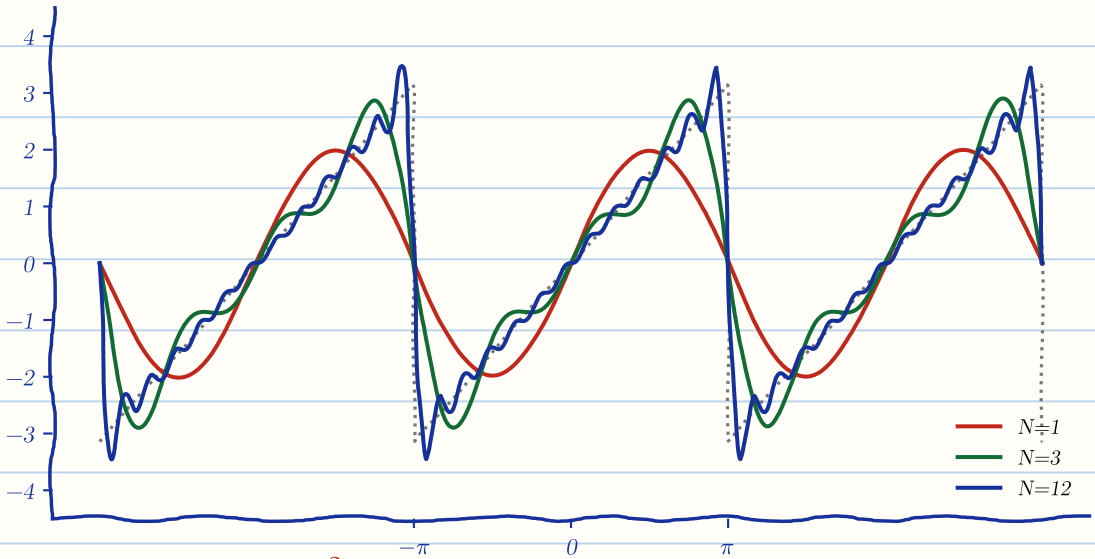
**Example 4.** $f(x) = x$ **in** $(-\pi, \pi)$ **(odd)**

$$b_n = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx = \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = \frac{2(-1)^{n+1}}{n}$$

$$x = 2\left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\right]$$

*Putting* $x = \frac{\pi}{2}$: $\frac{\pi}{2} = 2\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\right) \Rightarrow \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ *(Leibniz series).*

*Partial sums of the Fourier series of* $f(x) = x$




**Example 5.** $f(x) = x^2$ **in** $(-\pi, \pi)$ **(even)**

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2 dx = \frac{2\pi^2}{3},\quad a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos nx\,dx = \frac{4(-1)^n}{n^2},\quad b_n = 0$$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx = \frac{\pi^2}{3} - 4\cos x + \cos 2x - \frac{4}{9}\cos 3x + \cdots$$

*At* $x=\pi$: $\pi^2 = \frac{\pi^2}{3} + 4\sum\frac{1}{n^2} \Rightarrow \sum\frac{1}{n^2} = \frac{\pi^2}{6}$ ; *at* $x=0$:

$\sum\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$.

**Example 6.** $f(x)=|x|$ **in** $(-\pi,\pi)$ **(even)**

$a_0 = \frac{2}{\pi}\int_0^{\pi} x\,dx = \pi,\ a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{2[(-1)^n-1]}{\pi n^2}$ (*so* $a_n = -\frac{4}{\pi n^2}$ *for odd* $n$, $0$ *for even* $n$)

$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right]$$

*At* $x=0$: $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

**Example 7.** $f(x)=x^3$ **in** $(-\pi,\pi)$ **(odd)**

$b_n = \frac{2}{\pi}\int_0^{\pi} x^3\sin nx\,dx = 2(-1)^{n+1}\left[\frac{\pi^2}{n} - \frac{6}{n^3}\right] \Rightarrow$

$$x^3 = \sum_{n=1}^{\infty} 2(-1)^{n+1}\left[\frac{\pi^2}{n} - \frac{6}{n^3}\right]\sin nx$$

## 4. Fourier Series in an Arbitrary Interval

*For a function of period* $2l$ *defined in* $(c, c+2l)$ *put* $x \to \frac{\pi x}{l}$ :

**Interval (c, c+2l)**

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{l} + b_n\sin\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l}\int_c^{c+2l} f\,dx,\quad a_n = \frac{1}{l}\int_c^{c+2l} f\cos\frac{n\pi x}{l}\,dx,\quad b_n = \frac{1}{l}\int_c^{c+2l} f\sin\frac{n\pi x}{l}\,dx$$

- *On* $(-l,l)$: *even* $f \Rightarrow a_0 = \frac{2}{l}\int_0^{l} f,\ a_n = \frac{2}{l}\int_0^{l} f\cos\frac{n\pi x}{l},\ b_n = 0$; *odd* $f$ $\Rightarrow b_n = \frac{2}{l}\int_0^{l} f\sin\frac{n\pi x}{l}$.

**Example 8.** $f(x)=x^2$ **in** $(-l,l)$

$$\text{Even: } a_0=\frac{2}{l}\int_0^l x^2dx=\frac{2l^2}{3},\quad a_n=\frac{2}{l}\int_0^l x^2\cos\frac{n\pi x}{l}dx=\frac{4l^2(-1)^n}{n^2\pi^2}$$

$$x^2=\frac{l^2}{3}+\frac{4l^2}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos\frac{n\pi x}{l}$$

**Example 9.** $f(x)=2x-x^2$ **in** $(0,2)$ $(c=0,\ l=1)$

$$a_0=\int_0^2(2x-x^2)dx=\frac{4}{3},\quad a_n=\int_0^2(2x-x^2)\cos n\pi x\,dx=-\frac{4}{n^2\pi^2},$$

$$b_n=\int_0^2(2x-x^2)\sin n\pi x\,dx=0$$

$$2x-x^2=\frac{2}{3}-\frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{\cos n\pi x}{n^2}$$

*(The function is symmetric about $x=1$, which is why all sine terms vanish.)*

## 5. Half-Range Fourier Series

*A function given only on $(0,l)$ can be expanded in a series of SINES only or COSINES only (extend it as an odd or even function to $(-l,l)$):*

**Half-range expansions on (0, l)**

$$\text{Sine series: } f(x)=\sum_{n=1}^{\infty}b_n\sin\frac{n\pi x}{l},\qquad b_n=\frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$\text{Cosine series: } f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos\frac{n\pi x}{l}$$

$$a_0=\frac{2}{l}\int_0^l f\,dx,\qquad a_n=\frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$

**Example 10.** $f(x)=x$ **in** $(0,\pi)$ **: (a) cosine series (b) sine series**

$$(a)\ a_0=\frac{2}{\pi}\int_0^{\pi}x\,dx=\pi,\ a_n=\frac{2}{\pi}\int_0^{\pi}x\cos nx\,dx=\frac{2[(-1)^n-1]}{\pi n^2}:$$

$$x=\frac{\pi}{2}-\frac{4}{\pi}\left[\cos x+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\cdots\right]$$

(b) $b_n = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx = \frac{2(-1)^{n+1}}{n}$:

$$x = 2\left[\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\right]$$

**Example 11. Sine series of $f(x) = x(\pi - x)$ in $(0, \pi)$**

$b_n = \frac{2}{\pi}\int_0^{\pi} x(\pi - x)\sin nx\,dx = \frac{4[1-(-1)^n]}{\pi n^3}$ (*zero for even* $n$, $\frac{8}{\pi n^3}$ *for odd* $n$)

$$x(\pi - x) = \frac{8}{\pi}\left[\sin x + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \cdots\right]$$

**Example 12. Cosine series of $f(x) = \sin x$ in $(0, \pi)$**

$a_0 = \frac{2}{\pi}\int_0^{\pi} \sin x\,dx = \frac{4}{\pi}$ ; $a_n = \frac{2}{\pi}\int_0^{\pi} \sin x\cos nx\,dx = \frac{2[1+(-1)^n]}{\pi(1-n^2)}$ $(n \neq 1)$, $a_1 = 0$.

$$\sin x = \frac{2}{\pi} - \frac{4}{\pi}\left[\frac{\cos 2x}{1\cdot 3} + \frac{\cos 4x}{3\cdot 5} + \frac{\cos 6x}{5\cdot 7} + \cdots\right]$$

**Parseval's identity (extra)**

$$\frac{1}{\pi}\int_{-\pi}^{\pi}[f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)$$

# PART B : FOURIER TRANSFORMS

## 6. Fourier Integral Theorem (without proof)

*A NON-periodic function can be seen as the limit of a Fourier series when the period* $2l \to \infty$. *If* $f(x)$ *satisfies Dirichlet's conditions in every finite interval and* $\int_{-\infty}^{\infty}|f(x)|dx$ *converges, then*

**Fourier integral theorem**

$$f(x)=\frac{1}{\pi}\int_0^\infty\int_{-\infty}^\infty f(t)\cos\lambda(t-x)\,dt\,d\lambda$$

*At a point of discontinuity the left side is replaced by* $\frac{1}{2}[f(x^-)+f(x^+)]$.

**Fourier sine and cosine integrals**

$$f(x)=\frac{2}{\pi}\int_0^\infty \cos\lambda x\left[\int_0^\infty f(t)\cos\lambda t\,dt\right]d\lambda \quad (f \text{ even})$$

$$f(x)=\frac{2}{\pi}\int_0^\infty \sin\lambda x\left[\int_0^\infty f(t)\sin\lambda t\,dt\right]d\lambda \quad (f \text{ odd})$$

**Complex form of the Fourier integral**

$$f(x)=\frac{1}{2\pi}\int_{-\infty}^\infty e^{-i\lambda x}\left[\int_{-\infty}^\infty f(t)\,e^{i\lambda t}dt\right]d\lambda$$

**Example 13.** $f(x)=1$ **for** $|x|<1$, $0$ **for** $|x|>1$ **: express as a Fourier integral and deduce** $\displaystyle\int_0^\infty \frac{\sin\lambda}{\lambda}d\lambda$

$f$ *is even.* $\displaystyle\int_0^\infty f(t)\cos\lambda t\,dt=\int_0^1\cos\lambda t\,dt=\frac{\sin\lambda}{\lambda}.$

$$f(x)=\frac{2}{\pi}\int_0^\infty\frac{\sin\lambda\cos\lambda x}{\lambda}\,d\lambda$$

*Put* $x=0$ $(f(0)=1)$: $\displaystyle\frac{2}{\pi}\int_0^\infty\frac{\sin\lambda}{\lambda}d\lambda=1 \;\Rightarrow\; \int_0^\infty\frac{\sin\lambda}{\lambda}d\lambda=\frac{\pi}{2}.$

*At* $x=1$ *(jump): value* $\frac{1}{2}$*, giving* $\displaystyle\int_0^\infty\frac{\sin\lambda\cos\lambda}{\lambda}d\lambda=\frac{\pi}{4}.$

**Example 14. Fourier integral of** $f(x)=e^{-ax}$ $(x>0)$**, even extension** $e^{-a|x|}$

$$\int_0^\infty e^{-at}\cos\lambda t\,dt=\frac{a}{a^2+\lambda^2}\;\Rightarrow$$

$$e^{-a|x|} = \frac{2}{\pi}\int_0^\infty \frac{a\cos\lambda x}{a^2+\lambda^2}\,d\lambda \;\Rightarrow\; \int_0^\infty \frac{\cos\lambda x}{a^2+\lambda^2}\,d\lambda = \frac{\pi}{2a}e^{-a|x|}$$

## 7. Fourier Transform, Sine & Cosine Transforms

### Fourier transform pair

$$F(s) = F\{f(x)\} = \int_{-\infty}^{\infty} f(x)\,e^{isx}\,dx$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(s)\,e^{-isx}\,ds \qquad \text{(inverse transform)}$$

### Cosine & Sine transforms (and their inverses)

$$F_c(s) = \int_0^\infty f(x)\cos sx\,dx, \qquad f(x) = \frac{2}{\pi}\int_0^\infty F_c(s)\cos sx\,ds$$

$$F_s(s) = \int_0^\infty f(x)\sin sx\,dx, \qquad f(x) = \frac{2}{\pi}\int_0^\infty F_s(s)\sin sx\,ds$$

*(Some books use the symmetric form with factor $1/\sqrt{2\pi}$ (or $\sqrt{2/\pi}$); always follow the definition used in your question.)*

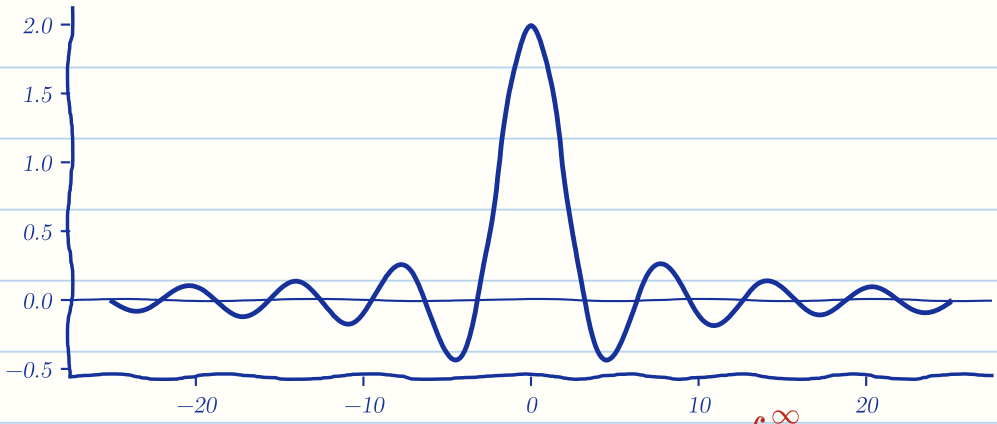
### Standard transforms (a > 0)

| $f(x)$ | $F_c(s)$ | $F_s(s)$ |
|---|---|---|
| $e^{-ax}$ | $\frac{a}{a^2+s^2}$ | $\frac{s}{a^2+s^2}$ |
| $1\ (0<x<a),\ 0\ (x>a)$ | $\frac{\sin as}{s}$ | $\frac{1-\cos as}{s}$ |
| $x\,e^{-ax}$ | $\frac{a^2-s^2}{(a^2+s^2)^2}$ | $\frac{2as}{(a^2+s^2)^2}$ |
| $\frac{e^{-ax}}{x}$ | - | $\tan^{-1}\frac{s}{a}$ |
| $e^{-a^2x^2}$ | $\frac{1}{2a}\sqrt{\pi}\,e^{-s^2/4a^2}$ | - |

**Example 15. Fourier transform of $f(x)=1$ for $|x|<a$, 0 for $|x|>a$**

$$F(s)=\int_{-a}^{a} e^{isx}\,dx=\frac{e^{isa}-e^{-isa}}{is}=\frac{2\sin as}{s}\quad (a\ \text{"sinc" curve}).$$

*Transform of the pulse $|x|<1$: $F(s)=2\sin s/s$*




**Example 16. Find $F_s$ and $F_c$ of $e^{-ax}$; hence $\displaystyle\int_0^\infty \frac{s\sin sx}{a^2+s^2}\,ds$**

$F_s=\displaystyle\int_0^\infty e^{-ax}\sin sx\,dx=\frac{s}{a^2+s^2}$. *Inverse:* $e^{-ax}=\dfrac{2}{\pi}\displaystyle\int_0^\infty \frac{s\sin sx}{a^2+s^2}\,ds\ \Rightarrow$

$$\int_0^\infty \frac{s\sin sx}{a^2+s^2}\,ds=\frac{\pi}{2}e^{-ax}.$$

**Example 17. Self-reciprocal: $f(x)=e^{-x^2/2}$**

$F(s)=\displaystyle\int_{-\infty}^{\infty} e^{-x^2/2}e^{isx}\,dx=\sqrt{2\pi}\,e^{-s^2/2}$ *(complete the square:* $-\frac{1}{2}(x-is)^2-\frac{s^2}{2}$*).*

**Example 18. Find $f(x)$ if $F_c(s)=\dfrac{1}{1+s^2}$**

*Since* $F_c\{e^{-x}\}=\dfrac{1}{1+s^2}$ *(take $a=1$),* $f(x)=\dfrac{2}{\pi}\displaystyle\int_0^\infty \frac{\cos sx}{1+s^2}\,ds=e^{-x}$.

## 8. Properties of Fourier Transforms

| Property | Statement $(F(s)=F\{f\})$ |
|---|---|
| *Linearity* | $F\{af+bg\}=aF(s)+bG(s)$ |

| Property | Statement $(F(s)=F\{f\})$ |
|---|---|
| Shifting in $x$ | $F\{f(x-a)\}=e^{isa}F(s)$ |
| Shifting in $s$ | $F\{e^{iax}f(x)\}=F(s+a)$ |
| Change of scale | $F\{f(ax)\}=\frac{1}{\lvert a\rvert}F(\frac{s}{a})$ |
| Modulation | $F\{f(x)\cos ax\}=\frac{1}{2}[F(s+a)+F(s-a)]$ |
| Derivative | $F\{f'(x)\}=-is\,F(s),\quad F\{f''\}=-s^2F(s)$ |
| Multiply by $x$ | $F\{xf(x)\}=-i\frac{d}{ds}F(s)$ |

### Cosine / sine transforms of derivatives

$$F_c\{f'(x)\}=sF_s(s)-f(0),\qquad F_s\{f'(x)\}=-sF_c(s)$$

$$F_c\{f''(x)\}=-s^2F_c(s)-f'(0),\qquad F_s\{f''(x)\}=-s^2F_s(s)+s\,f(0)$$

*(valid when* $f, f'\to 0$ *as* $x\to\infty$*.) Also*

$F_c\{f(x)\cos ax\}=\frac{1}{2}[F_c(s+a)+F_c(s-a)]$, *and*

$F_s\{f(x)\cos ax\}=\frac{1}{2}[F_s(s+a)+F_s(s-a)]$.

*Example:* $F_c\{xe^{-x}\}$ *via* $F_c\{xf\}=\frac{d}{ds}F_s\{f\}$: $\frac{d}{ds}\frac{s}{1+s^2}=\frac{1-s^2}{(1+s^2)^2}$ *(agrees with table,* $a=1$*).*

## 9. Convolution Theorem & Parseval's Identity

*Convolution of* $f, g$ *on* $(-\infty,\infty)$: $(f*g)(x)=\int_{-\infty}^{\infty}f(u)\,g(x-u)\,du.$

### Convolution & Parseval

$$F\{f*g\} = F(s)\cdot G(s) \quad \Longleftrightarrow \quad f*g = F^{-1}\{F(s)G(s)\}$$

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(s)|^2 ds \qquad \text{(Parseval)}$$

$$\int_0^{\infty} F_c(s)G_c(s)\,ds = \frac{\pi}{2}\int_0^{\infty} f(x)g(x)\,dx$$

**Example 19. Evaluate $\displaystyle\int_0^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)}$ using Parseval**

*Take* $f = e^{-ax}$, $g = e^{-bx}$: $F_c = \dfrac{a}{s^2+a^2}$, $G_c = \dfrac{b}{s^2+b^2}$. *Then*

$$ab\int_0^{\infty} \frac{ds}{(s^2+a^2)(s^2+b^2)} = \frac{\pi}{2}\int_0^{\infty} e^{-(a+b)x}dx = \frac{\pi}{2(a+b)}$$

$$\int_0^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2ab(a+b)}$$

**Example 20. Use Parseval to evaluate $\displaystyle\int_0^{\infty} \frac{\sin^2 x}{x^2}dx$**

*Pulse* $f = 1$ *on* $(0,1)$: $F_c = \dfrac{\sin s}{s}$. $\displaystyle\int_0^{\infty} F_c^2 ds = \frac{\pi}{2}\int_0^{1} 1\,dx = \frac{\pi}{2} \Rightarrow$

$$\int_0^{\infty} \frac{\sin^2 s}{s^2}ds = \frac{\pi}{2}.$$

## 10. Quick Revision Sheet

**Formulae at a glance**

$2\pi$-periodic: $a_0 = \frac{1}{\pi}\int f$, $a_n = \frac{1}{\pi}\int f\cos nx$, $b_n = \frac{1}{\pi}\int f\sin nx$ (series $\frac{a_0}{2} + \ldots$).

Even $f$: only $a_n$ (with $\frac{2}{\pi}\int_0^{\pi}$); odd $f$: only $b_n$. Period $2l$: replace $nx \to \frac{n\pi x}{l}$, $\frac{1}{\pi} \to \frac{1}{l}$.

Half range on $(0, l)$: $\frac{2}{l}\int_0^{l}$ with $\sin\frac{n\pi x}{l}$ (sine) or $\cos\frac{n\pi x}{l}$ (cosine).

At a jump the series gives the AVERAGE of left and right limits.

$F(s) = \int f e^{isx}dx$; $F_c = \int_0^{\infty} f\cos sx$, $F_s = \int_0^{\infty} f\sin sx$; inverses have factor $\frac{2}{\pi}$ (sine/cosine) or $\frac{1}{2\pi}$ (complex).

- Always check even/odd symmetry FIRST - it saves half the work. Quote Dirichlet conditions when asked "existence".
- Typical long questions: Fourier series of a piecewise function, half-range expansions, Fourier integral deduction ($\int \sin\lambda/\lambda$), sine/cosine transform of $e^{-ax}$, Parseval/convolution evaluation.